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DermaScanAI an explainable hybrid deep learning framework for automated skin lesion classification using dual attention and metadata fusion

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Skin cancer is more common and can be fatal if not diagnosed and treated promptly. Automated skin lesion classification based on dermoscopic images has attracted significant attention, especially with the rapid rise of deep learning-based methods. Yet, it still suffers from limitations, including insufficient multi-scale representation, a lack of global context modelling, and less effective feature recalibration, which undermine classification reliability and robustness. Furthermore, the lack of transparency associated with many deep learning models erodes trust and acceptance among healthcare providers. To tackle these issues, we propose a new deep learning framework that combines multi-scale convolutional feature extraction with lightweight Transformer encoders, while incorporating squeeze-and-excitation (SE) blocks to improve channel-wise attention. By exploiting spatial granularity, contextual richness, and adaptive feature recalibration across latent class-specific patterns in dermoscopic images, the proposed model classifies images from the HAM10000 dataset into seven skin lesion categories. It uses a multi-stage architecture to embed both local and global patterns, and Grad-CAM as a post-hoc explainability method to promote interpretability. We conduct experimental evaluations on the publicly available HAM10000 dataset, and the results indicate that the proposed strategy achieves competitive performance with the state-of-the-art methods, reaching 94.8% overall accuracy, 91.9% macro-average F1-score, and 0.957 AUC. Class-wise performance analysis and ROC curves demonstrate robustness across a range of lesion categories, while ablation experiments verify the individual contributions of each architectural component. In conclusion, our framework outlines a possible computational approach that combines interpretation with noise resilience to refine automated classification of skin lesions. However, further prospective, multi-institutional validation will be necessary before it can have implications for future clinical decision support systems.

Keywords Skin cancer detection, Deep learning, Transformer networks, Attention mechanisms, Medical image classification

Skin cancer remains the most common malignancy in the world; if diagnosed late, melanoma develops into one of the most aggressive and lethal variants of skin cancer. As skin cancer rates are increasing and the number of experienced healthcare professionals is decreasing in many regions, CAD (Computer-Aided Diagnostic) systems are considered especially needed. Artificial intelligence and deep learning approaches have recently proven to be high-performance tools for analysing medical images, with skin lesion classification and diagnosis exhibiting high accuracy. So far, this has led convolutional neural networks (CNNs) to outperform all other approaches for extracting high-level features from dermoscopic images and classifying skin conditions[1,2]. Nevertheless, current methods suffer from (1) a small dataset leading to overfitting, (2) no provision for lesion-specific feature exploration, and (3) no combination of spatial and contextual information. To further improve performance[3], hybrid architectures combining CNNs with attention- or Transformer-based encoders have been studied in several works. However, these methods ignore the multi-scale information required for segmenting

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fine-grained lesion boundaries, or they do not address the class imbalance common in datasets like HAM10000[4], which is necessary for better segmentation. Even more importantly, most of these works ignore the need for sufficient interpretability to justify model predictions, which is crucial for acceptance in real-world clinical practice. Here, we present a study to bridge these research gaps by offering a new deep learning framework that incorporates traditional multi-scale convolutions, Transformer blocks for capturing global dependencies, and squeeze-and-excitation (SE) modules for facilitating channel-wise attention. This framework aims to improve classification precision and enhance intelligibility through explainability practices, such as Grad-CAM. In this work, we propose a trade-off solution: an accurate, interpretable, and scalable skin cancer classification model that achieves competitive accuracy on standardised datasets and integrates well with clinical workflows for skin cancer diagnosis.

Hence, this study aims to develop a hybrid deep learning model that effectively integrates local, global, and channel-wise features to classify dermoscopic images into seven clinically distinct skin cancer categories accurately. Its significant contributions include multi-scale spatial pyramidal pooling (MSPP), which extracts features at different spatial resolutions; Transformer encoders that learn global features; spatial squeeze-and-excitation (SE) blocks to capture channel dependencies; and Grad-CAM to provide post-hoc visual explanations for attendees' predictions. This motivated us to introduce a holistic set of techniques that capture all the ways the disadvantages of existing models are mitigated, to progress towards a new level of precision and transparency in automatic skin cancer diagnosis.

The significant contributions of this paper are summarised as follows. We propose a new multi-scale attention-guided deep learning model for context-aware classification of lesions, with a unified architecture of CNN, Transformer, and SE modules, to outperform the state of the art. Second, the framework is evaluated using the HAM10000 dataset, and its exhaustive performance is reported. In the third analysis, it proceeds to a confusion matrix and ROC-based analysis, showing the model's class-wise robustness and the accessibility of different types of skin lesions. Fourth, we perform an ablation study on other modules in the proposed architecture. Fifth, we use Grad-CAM to make our results more explainable and thus more relatable to clinicians. Finally, experimental results comparing the proposed system with recent state-of-the-art models highlight its benefits in both performance and generalisation.

The paper is organised as follows. Section 2 offers a comprehensive literature review that highlights gaps in existing research and situates this work within previous studies. The theoretical basis and related preliminaries for model development are given in Sect. 3. In Sect. 4, we explain our proposed method, including the architecture, training process, and reasoning for design choices. Section V provides the experimental results, including dataset description and evaluation metrics, performance analysis, comparative studies, ablation experiments, and explainability results. In Sect. 6, we discuss our findings in detail and explain the limitations of our study. At last, Sect. 7 wraps up the paper and outlines future research directions to address challenges in generalisation, deployment, and integration into clinical practice.

Related work

Deep learning, explainable AI, and hybrid modelling are among recent advances in artificial intelligence that have improved skin cancer detection. We classify the existing literature into five main research themes to showcase technological advances, assess model capabilities, and identify research gaps that our proposed work seeks to address. In each subsection, we cluster relevant studies according to methodological aspects and application domain .

Deep learning techniques for skin cancer detection

Deep learning models, such as CNNs, reveal exceptional promise for the classification and segmentation of skin lesions. Subsequent techniques have been investigated in multiclass classification, dermoscopic image analysis, and advancements in adversarial learning to improve generalisation across diverse datasets.

Tahir et al. However, in[1], the authors introduced DSCC_Net, a strong multi-classifier that achieved good performance in skin lesion classification. A field report that compared the advantages and disadvantages of using machine learning and deep learning models in clinical contexts[2], a conclusion further confirmed by Bechill et al. (Anonymous Author(s) under preparation). Nawaz et al. Derouiche et al.[5] presented a CNN architecture for lesion detection that improved the state of the art in this domain, demonstrating its application in analysing dermoscopic images. Shetty et al. For example[6], presented a solution that combines CNNs and classical machine learning methods to achieve accurate classification of skin lesions. Collectively, these studies emphasise existing attempts that use hybrid architectures and augmentation modules at either the pre- or post-processing stage to advance dermatology diagnostics. Ali et al. Enhanced early diagnosis, Abunadi and Senan[7], combines ASSANDRA evaluation, which does not scale with small data[8].

To enhance early diagnosis, Abunadi and Senan[7] combined ML and DL approaches, focusing on preprocessing and segmentation. Wu et al.[9] A systematic review of DL methods for skin cancer classification reported that CNNs were the most widely used architecture and that Ensemble Learning approaches were increasing in use. Shah et al. Herein[10], used CNNs for feature selection and ML-based classification, applying explainable AI mechanisms. Giavina-Bianchi et al. Finally[11], addressed the relevance of clinical trust in AI systems by comparing several visual explanation methods for melanoma diagnosis and their agreement with dermatologists. De Souza Jr. et al. For instance[12], used CNNs for Barrett's oesophagus cancer detection, further providing explanations for predictive decisions, and this technique is also helpful for skin-image-based cancer analysis.

Nigar et al. Instead, a deep learning-based model is presented in[13], which combines with XAI for skin lesion classification, achieving improvements in transparency and accuracy in multiclass settings. Ahmad et al. The work of[14] proposes a multiclass skin lesion classification framework using DL and XAI to address two clinical concerns, namely class imbalance and clinical trust. A genetic algorithm-based explainer for melanoma images was described by[15], integrating auto-segmentation and XAI to perform global diagnosis. In 2019, Yasir and Kim[16] introduced a deep feature aggregation network that employs simple attention mechanisms for enhanced lesion representation. Shaik et al. This study[17] combined CNNs and BiLSTMs with attention layers to effectively classify lesions, capturing both spatial and temporal dependencies.

While Alahmadi[18] developed a U-Net to accurately extract skin boundaries. Liu et al. proposed a VAN-based cross-attention mechanism in[19] to improve multiscale lesion segmentation accuracy in noisy environments. Wang et al.[20] used adversarial multimodal fusion with attention to combine clinical and dermoscopic images within the same modality, thereby improving contextual information for lesion classification. In an experimental work, Gulzar and Khan[21] comparatively analysed CNNs and Vision Transformers and demonstrated their performance and computational cost trade-off. Abdelhalim et al. To augment skin lesion data to introduce diversity and improve model robustness[22], employed a self-attention-based progressive GAN (He et al.). Deep metric attention learning for lesion classification improves class separability and interpretability in dermoscopic analysis.

Hossain et al. A first ensemble framework that uses max voting over pre-trained DL models to increase skin cancer[23] detection accuracy. Jaiyeoba et al. And[24] presented a systematic review of AI-based methods for skin disease detection, focusing on dataset diversity, evaluation metrics, and clinical limitations. Nancy et al.[25] conducted a comparative analysis of ML and DL and reported the advantages of using CNNs for skin cancer diagnosis. Ghosh et al. Deep CNN-based architectures are superior to classical models across datasets, while DL and ML are overall competitive[26]. Gamil et al. Improved classification performance by an optimised AdaBoost model for skin cancer detection[27],

Midasala et al. MFEUsLNet is a multilevel feature extraction model based on unsupervised learning that detects features more confidently and addresses the limitations of angular-based models, as presented in Ref[28]. At an early stage, Mahmoud and Soliman[29] developed a hybrid novel AI-based diagnosis system to enhance prediction accuracy. Natha et al. Their solution was an ensemble-based methodology that leverages multiple base learners to improve classification performance[30]. Saeed et al. For example[31], proposed a novel augmentation-based DL pipeline that leverages generative AI to enhance skin cancer prediction accuracy. Musthafa et al. To improve lesion classification accuracy, Rajasharma et al. proposed an optimised convolutional neural network with a checkpoint mechanism to curb overfitting for diagnosing skin cancer[32].

Explainable and interpretable AI models in dermatology

Explainable AI (XAI) is becoming increasingly commonplace in dermatology to mitigate the "black-box" problem of deep learning. Gradient-based approaches, such as Grad-CAM, and perturbation-based methods, such as LIME or interpretable transformers, contribute to making the model's decisions interpretable and to driving transparency and clinical prediction.

Grignaffini et al. The summary[3] included a broad systematic review of ML methods and highlighted issues such as accuracy and generalisation. Naeem et al. Ejaz et al.[5] used SCDNet, a convolutional network with residual blocks, to identify skin cancers in compiler images. Gururaj et al. DeepSkin[33] was developed based on transfer learning and a dense network, with consideration of class imbalance and high sensitivity. Nawaz et al. In[34], they used deep learning with fuzzy k-means clustering for better segmentation. Kaur et al. developed a melanoma classification method based on an innovative CNN (base network) model[35], which used high-pass filtering to enable a conventional CNN to outperform other methods. Venugopal et al. Next, an EfficientNet of[36] was employed to offer a modified solution for a fully automatic and robust dermoscopic analysis.

Proposed by Mehr and Ameri[37], which is especially formulated for image-level classification in a robust manner over optimised architectures. Naeem et al. Similarly[38], proposed the SNC_Net, which combines classic feature extraction with deep learning outputs to achieve better performance. Grignaffini et al. Morid et al.[39] introduced a melanoma diagnosis framework utilising CNNs, incorporating feature injection and explainability modules. Mridha et al. Nesarale et al.[40] We have proposed an interpretable skin cancer model based on an optimised CNN, utilising the innovative healthcare system. Gamage et al. The work in[41] presents a mask-guided, explainable CNN model for melanoma detection, with accurate lesion localisation and visual explanations of its decisions.

Ali et al. proposed a hybrid model, xCViT, that combines ViTs, CNNs, and the Xception module[42] for interpretable skin disease detection. Dagnaw et al. Vision Transformers equipped with XAI mechanisms were utilised in[43] and demonstrated suitability for skin cancer tasks. A hybrid framework of RBF-Net + CNN with explainability for reliable diagnosis was proposed by Ullah and Zia[44]. Liu et al. An XAI-based pipeline for predicting skin cancer risk from facial features was designed by[45] for non-invasive skin cancer screening. Behara et al. Grid-Based dimensionality analysis and interpretation for CNNs[46].

Chanda et al. This is clearly manifested in increased trust and diagnostic accuracy in melanoma detection with dermatologist-like explainable AI models[47]. Nie et al. Therefore[48], developed ELA-Net, an attention-based model with optimisations to expedite medical care in low-resource environments. Zhuang et al. To address class imbalance in medical image recognition[49], proposed a class-attention mechanism tailored to lesion regions. Manzari et al. BEFUnet[50] fused CNN and Transformer layers to perform accurate medical segmentation. Takiddin et al. A scoping review of artificial intelligence applications in skin cancer detection: trends, limitations and research opportunities, conducted by[51].

Bhatt et al. In[52], the authors surveyed the most advanced ML techniques for melanoma detection and evaluated the performance of CNN, SVM, and hybrid models. Alzamili and Ruhaiyem[53] provided a high-level overview of challenges and research gaps in early skin cancer diagnosis using DL/ML, with promising future directions. Behara et al. A review by[54] comprehensively discusses the role of AI in dermatological diagnosis, specifically skin cancer detection. Rezk et al.[55] explored different AI approaches to the diagnosis and treatment of skin cancer, describing technical challenges and potential solutions. Melarkode et al.[56] provided a detailed review of the state of the art in AI-based skin cancer detection and highlighted several unmet challenges, including dataset bias and interpretability. Abou Ali et al. Many studies have developed high-precision DL models for tumour classification, such as a model that combines image enhancement and classifier fusion to classify skin lesions[57] reliably.

Attention mechanisms and vision transformers for skin lesion analysis

Specifically, attention-based methods and transformer architectures were attacked and adapted to achieve practical model focus and overall performance, alongside ViT, TransUNet, and ELA-Net. Such methods enhance the contextual representation of lesion characteristics, applicable to both segmentation and classification.

Khater et al. Fang et al.[58] reviewed post-hoc XAI techniques for skin cancer classification from pre-extracted image features, aiming for a transparent, evidence-based embrace of AI. To address the class imbalance problem in skin cancer classification, where certain malignancy types are rare, Ravi[59] proposed an attention-based cost-sensitive deep learning model to improve sensitivity and precision. Naveed et al. AD-Net: A stack of attention and dilated convolutional residual blocks with a guided decoder for biomedical image segmentation[60]. The works by Pedro and Oliveira[61] analysed the effects of attention and self-attention mechanisms on lesion classification and demonstrated the power of attention in filtering relevant features. Younas et al. An Inception-Residual CNN with Attention Layers for Skin Cancer Diagnosis[62].

Noor et al.[63] employed dataset refinement and attention-based vision modules to improve classification robustness in clinical datasets. Nie et al. To address this challenge[48], proposed an attention-based model, ELA-Net, optimised for low-resource medical environments. Zhuang et al. Class imbalance in medical image recognition is addressed using a class-attention mechanism dedicated to lesion regions[49]. Manzari et al. BEFUnet swapped some of the U-Net's CNN layers for Transformers and achieved highly dense medical segmentation[50]. Sharma et al. An evaluation in the context of medical classification has been performed for Vision Transformers and CNNs[64], where performance gains and complexity trade-offs are discussed.

Bougourzi et al. In[65], a D-TrAttUnet architecture was proposed, a hybrid architecture that captures both subtle and generic medical features. Kim et al. Explainability, Accuracy, and Integration in Clinical Workflows[66]. Wen et al. In a systematic review conducted by[67], the authors identified the quality, diversity, and usability of skin cancer image datasets and emphasised the need for standard and diverse datasets (Nguyen et al.). In contrast with[68], who compared ML algorithms for melanoma classification, they observed that ensemble models outperformed single decision trees, SVMs, and k-NN.

Hybrid CNN–CNN-transformer architectures in medical imaging

New hybrid architectures that combine CNNs with transformers have been proposed to extract local and globally relevant features. They provide more accurate segmentation, classification, and other stroke lesion detection across various medical modalities.

Djoumessi et al. This balance was proposed by[69] to provide a trade-off between accuracy and explainability. Hu et al. An SSM architecture that fused CNN and Transformer-based layers was proposed in[70] for a more robust medical image classification. Zeynali et al. Xception-based Feature Refinement of a Hybrid CNN-Transformer Model for Improved Breast Cancer Classification[71] Li et al. TFCNs: In 2020[72], proposed a hybrid network that combines CNNs and Transformers for medical image segmentation and achieved improvements in maintaining anatomical features. Lan et al. developed a U-shaped hybrid CNN-Transformer model for precise segmentation of biomedical images, called Brau-Net++[73].

Vindas et al. proposed a hybrid model with multi-feature extraction and attention fusion to classify cerebral embolism in[74] precisely. Singh[75] used a CNN-Transformer hybrid model to extract features from chest X-ray images for thoracic disease diagnosis and reported high classification accuracy. Kuang et al. Gu et al.[76] proposed a new CNN-Transformer-based methodology, grounded in the circular feature interactions framework, for acute stroke lesion segmentation in non-contrast CT. ConvFormer[77] proposed to combine convolution and transformer layers and achieve better performance than many other methods in the medical imaging benchmarks. Bi et al. An interactive hybrid model for musculoskeletal imaging was recently proposed[78], achieving better diagnostic performance by deep-fusing local and contextual features. Iqbal et al. We propose a hybrid deep learning model, TBConvL-Net, that can improve segmentation performance on complex medical images[79].

In biomedical image analysis, recent studies have begun to investigate hybrid deep learning architectures that leverage attention mechanisms, transformer models, and explanations to improve the reliability of results. Banerjee et al. Only two papers leverage transformers for brain tumor classification[80]: demonstrates the enhanced robust and generalizable nature of pyramidal attention and transfer learning for the brain tumor classification task, highlighting the importance of clinically reliable AI design; Building on this orientation for dermatology, Banerjee et al. A complete comparison of existing skin cancer detection techniques using bipartite convolutional and attention-based methods was reported by[81], revealing a similar trend towards explainable AI and transfer learning in dermoscopy. In[82,83], broader perspectives on hybrid deep learning paradigms are presented, where the overall performance of transformer-based, kernel-assisted models is systematically evaluated on diabetic retinopathy detection tasks and their ability to capture complex visual patterns from occupied images. Attention-based methods handle class imbalance and the complexity of visual patterns, achieving competitive performance, as shown in[82,83]. In complementary work, Sujini[84] presented a hybrid Vision Transformer–WGAN framework for classifying thyroid ultrasound imagery, demonstrating that transformer-centric designs can be versatile across imaging modalities. Together, these works support the pertinence of attention-guided hybrid models for interpretable and generalizable medical image classification.

Surveys, datasets, and future directions in AI-based dermatology

Many survey papers and/or scoping reviews have summarised datasets, trends in recent years, benchmarking and research gaps of dermatology-Champions-supported through AI-based tools for skin cancer diagnosis. Such works are a valuable roadmap for methodological and clinical translation moving forward.

Takiddin et al. A scoping review of AI applications for the diagnosis of skin cancer: trends, limitations, and research opportunities[51]. Jaiyeoba et al. A comprehensive review of the existing AI-based skin disease detection techniques with respect to dataset diversity, performance metrics, and diagnostic limitations[24]. Bhatt et al.[52] reviewed cutting-edge ML melanoma detection techniques and evaluated CNN, SVM, and hybrid models. Nancy et al. conducted a comparative analysis of machine learning and deep learning algorithms and found that CNNs are state-of-the-art for the diagnosis of skin cancer[25]. Wen et al.[67] Quality, diversity and reproducibility of skin cancer image datasets.

Alzamili and Ruhaiyem[53] provided an extensive survey, highlighted the challenges and gaps in DL/ML-based early-stage skin cancer diagnosis, and suggested potential directions for the future. Behara et al. provide a thorough and informative review of the changing role of AI in dermatological diagnosis[54], specifically in skin cancer detection. Nguyen et al. Performed comparative research on ML algorithms[68]. ML-based algorithms for melanoma classification showed the best performance with ensemble models. Rezk et al. While more theoretical in nature[55], also described the technical limitations and opportunities of several artificial intelligence strategies for skin cancer diagnosis and management. Melarkode et al. AI-Assisted Skin Cancer Diagnosis: A Review of Current Advances and Open Challenges. We address recent advancements in AI-powered skin cancer diagnosis and categorise them by application area.

Table 1Comparison of eight closely aligned studies in terms of findings in deep learning-based skin cancer detection. The first two studies primarily focus on hybrid approaches, attention mechanisms, and XAI methods to improve diagnostic accuracy. Common aspects of these studies, for example, (Ali et al[42]. and Musthafa et al.). In[32], a hybrid CNN–Transformer architecture was proposed, while a CNN-optimised architecture was described in[85]to improve classification accuracy—Djoumessi et al. Zeynali et al[69]. et autres (2016). Two-stage architectures that integrate convolutional and transformer features for coordination-based medical image analysis —[71]. Although we have made remarkable progress, there are still limitations, such as the lack of XAI and real-time classifiability, as well as robustness to multiple lesion types and imaging conditions. Inspired by these observations, we introduce a new hybrid architecture that captures information at multiple spatial scales through a dual-attention mechanism, resulting in a model that is both accurate and interpretable and that overcomes the hallmark limitations of state-of-the-art frameworks: scalability, clinical trustworthiness, and sensitivity to lesion variability.

Preliminaries

In this section, we describe the background knowledge and technologies that underpin the proposed LungCare-AI framework. It provides essential background on convolutional neural networks (CNNs), attention mechanisms, and squeeze-and-excitation (SE) modules, which form the theoretical foundation for the model architecture. This section clarifies the deep learning elements used in the method.

Background of skin cancer diagnosis using deep learning

Skin cancer is one of the most prevalent types of cancer worldwide, and melanoma, the most aggressive subtype, due to its ability to disseminate. Improved sensitivity and specificity in early diagnosis are prerequisites for developing effective therapeutic remedies and preventing patient deaths. Dermoscopy, or dermoscopic imaging, is a non-invasive, widely used, and active approach that maximises the quality of imaging of skin lesions to

Ref	Authors (year)	Methodology	Dataset used	Key findings	Research gap
42	Ali et al. (2025)	xCViT: CNN-Xception-ViT hybrid with XAI	HAM10000	Achieved high accuracy in skin disease classification with explainability	Lacks multi-scale feature fusion and spatial channel attention
43	Dagnaw et al. (2024)	Vision transformer with XAI	ISIC 2018	The transformer-based model achieved good classification accuracy while improving interpretability.	A transformer cannot extract strong low-level features from a CNN alone.
47	Chanda et al. (2024)	Dermatologist-level explainable AI	ISIC 2019	Enhanced clinician trust and diagnostic confidence using transparent predictions	No hybrid deep learning model or attention-based refinement
17	Shaik et al. (2025)	Attention-based CNN-BiLSTM hybrid	ISIC 2020	Improved lesion classification by leveraging spatial-temporal features	Lacks explainability and does not include SE-based channel recalibration
20	Wang et al. (2022)	Multimodal fusion using clinical and dermoscopic images with attention	ISIC + clinical data	Combined modalities improved accuracy over single-source models	High model complexity and poor scalability for edge deployment
50	Manzari et al. (2024)	BEFUnet: CNN-Transformer hybrid segmentation network	Generic medical images	Achieved precise segmentation with transformer integration	Not focused on skin lesions; lacks skin-specific explainability modules
65	Bougourzi et al. (2024)	D-TrAttUnet: Transformer + CNN for segmentation	ISIC 2018, PH2	Effective for subtle lesion segmentation in dermoscopy images	Does not integrate multi-scale attention or SE modules
71	Zeynali et al. (2024)	CNN-Transformer with Xception-enhanced features	Breast cancer dataset	Achieved high classification performance in a non-skin domain	Not dermatology-specific; lacks relevance to dermoscopic lesion detection

Table 1. Comparative summary of related deep learning-based skin cancer detection studies highlighting methodologies, datasets, key findings, and research gaps addressed.

facilitate detailed visual examinations by dermatologists. Despite the widespread use of dermoscopic images to improve diagnostic accuracy, image evaluation is performed manually, making it subjective, time-consuming, and dependent on the clinician, leading to variability in diagnostic results.

In response, deep learning has emerged in medical image analysis and achieved breakthroughs across many automated disease-detection tasks, thereby circumventing the limitations noted above. In particular, CNNs have performed excellently in feature extraction and classification across various medical imaging modalities, including skin lesion classification. ResNet, DenseNet, and Inception are examples of modern CNN architectures that have achieved high accuracy in melanoma versus non-melanoma lesion classification on dermoscopic images. However, newer network architectures learn their feature representations directly from the data, hierarchically, rather than the previous methods, which involved human expert work in feature engineering.

While they achieve decent performance, conventional convolutional neural network (CNN) architectures are naturally designed to capture spatial information within a local region and do not adequately exploit global context within dermoscopic images, which is essential for an accurate understanding of lesions. Moreover, these models are trained only on visual information. They therefore cannot utilise clinical metadata, including patient age, sex, and the anatomical site of the lesion, which are critical for diagnostic interpretation. The black-box nature of DL models, which is difficult to trust and interpret[35], challenges the imperative use.

Recent work in medical imaging has incorporated attention mechanisms and transformer-based models to close such gaps in feature extraction by selectively emphasising relevant image regions and capturing long-range dependencies. When combined with explainable AI methods, these methods constitute a step towards more transparent and clinically interpretable decisions in deep learning models. Inspired by these advances, we present an explainable, high-performance skin cancer detection framework that combines a convolutional and transformer architecture, incorporating dual-attention and metadata fusion mechanisms to achieve the proposed goal.

Existing deep learning architectures for skin cancer detection

Deep learning models have addressed the skin lesion classification problem by performing end-to-end automated feature extraction and classification. Throughout these developments, some architectures have become more common than others, such as CNNs, which have demonstrated outstanding performance in numerous medical image-based tasks. VGGNet, ResNet, DenseNet, and Inception architectures are classical CNNs that are often used for their ability to derive hierarchical representations from dermoscopic images. They employ a combination of convolutional layers, pooling layers, and non-linear activations to generate feature-space representations that capture texture, shape, and colour patterns of lesions. Mathematically, we define the convolution operation, which is the core operation performed by a CNN layer, as in Eq. 1.

$$X^{(l)} = f\left(W^{(l)} * X^{(l-1)} + b^{(l)}\right) \tag{1}$$

where $X^{(l-1)}$ and $X^{(l)}$ respectively are the input and output feature maps for the l-th layer, $W^{(l)}$ is the convolution kernel, $b^{(l)}$ is a bias term, and $f(\cdot)$ is a non-linear activation function— typically u(t)=maxt) (a Rectified Linear Unit, ReLU).

While effective, convolutional neural networks (CNNs) primarily extract local spatial features owing to the small field of view of convolutional layers. Such a shortcoming limits their capability to learn global contextual information across the entire dermoscopic image, which is necessary for correctly classifying two or more skin lesions that often appear visually similar. Moreover, CNN-based models typically treat each image as an independent sample and do not utilize additional clinical information (e.g., patient demographics or lesion location) that could enhance diagnostic performance.

Transformer-based models have recently emerged in the medical imaging field to address these constraints. Transformers were initially developed for sequence modeling in the field of natural language processing. Vision Transformers (ViT) and its derivatives apply this method to image data by dividing the image into patches, which are then processed as a sequence of tokens. In this way, the self-attention mechanism in transformers computes the relationship between image patches, as shown in Eq. 2.

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

Where Q, K, and V are the query, key, and value matrices from the input features, and d_k is the dimensionality of key vectors. This allows the model to give different importance when making predictions.

While transformers work well in the global context, large-scale datasets and high computational resources are required to train most transformer-based models, which hampers their direct application to medical datasets like HAM10000. To alleviate this, recent works propose hybrid architectures that utilize convolutional layers to extract local features[48] or incorporate transformer encoders to capture long-range, global spatial dependencies[26], thereby balancing the needs for computational efficiency and contextual modeling.

In addition, existing works rarely consider patient metadata as part of the classification pipeline. Clinical metadata, including age, sex, and anatomical site, provides crucial clinical context but is frequently segregated or completely ignored during processing. Such a loss hampers the real-world application of deep learning models in diagnosis.

The current work builds upon these innovations by proposing a hybrid DL framework with a ConvNeXt-based convolutional backbone, two attention modules, and a lightweight transformer encoder to capture local and global features. In particular, it combines patient metadata at the feature level to enhance the context-aware nature of the classifier, thereby overcoming earlier challenges faced by solely CNN or transformer-based models.

Problem definition

The research presented in this paper is primarily aimed at formulating the task of skin cancer detection as a multi-class classification problem in which the system classifies of several classes. The images are also associated with optional patient metadata (age, sex, and anatomical site of the lesion), providing crucial clinical context. Introduction. While several systems exist that utilize only visual information to generate a diagnosis, other systems start to exploit clinical information in follow-up diagnoses, which results in a trade-off between visual explainability and clinical relevance.

Denote $I \in \mathbb{R}^{H\times W\times 3}$ the dermoscopic image with spatial resolution as $H \times W$, and the $M \in \mathbb{R}^k$ metadata vector $k$with attributes from, i.e., clinical attributes of the patient We attempt to learn a classification function $f(\cdot)$ parameterized by Θ, mapping the image and metadata features to a label $y \in \{1,2,\ldots,C\}$, where C is the total number of lesion types. We can write this mathematically as in Eq. 3.

$$y = f(I, M; \Theta) \tag{3}$$

The classification model needs to capture both local and global image features, representing lesion-specific characteristics. Clinical context that might change the features of a lesion. There are some challenges that need to be solved while learning the function $f(\cdot)$.

Feature extraction: Dermoscopic images have large intra-class variablility and inter-class similarity requiring feature extraction mechanisms that can differentiate high-dimensional features among the lesion types.

Metadata Fusion: Clinical metadata must be fused with image-derived features for context-aware classification rather than processed in isolation or completely ignored.

Class Imbalance: Since the lesion classes are not proportionately distributed and melanoma is a minority class, the classification should de-bias the classifier from the majority class using a proper loss function like focal loss (see Eq. (1)).

Interpretability: Outputs must be interpretable, relying on explanation AI techniques to provide explanations that clinicians can use to base their classification rationale on. It means visual explanation using Grad-CAM + + and feature importance insights from SHAP analysis.

More formally, the training process aims to minimize the classification loss $\mathcal{L}_{cls}$, defined over a dataset $D = \{(I_i, M_i, y_i)\}_{i=1}^{N}$, where N is the number of training samples, by optimizing the model parameters Θ as in Eq. 4.

$$\Theta^* = arg\min_{\Theta} \frac{1}{N} \sum {}_{i=1}^{N} \mathcal{L}_{cls}(f(I_i, M_i; \Theta), y_i) \tag{4}$$

Classification loss $\mathcal{L}_{cls}$ we use focal loss to tackle class imbalance problems as introduced before (Eq. 1).

Few studies have exploited the promising hybrid deep learning frameworks that combine convolutional and transformer-based architectures for skin cancer detection to their full potential, while also incorporating novel dual attention mechanisms and clinical metadata fusion for explainable predictions.

Proposed framework

This section presents the LungCare-AI architecture designed for accurate lung cancer detection and classification. It integrates Convolutional layers that serve for multi-scale feature extraction, along with attention and SE modules, to enhance representation learning. The section details each component of the pipeline, including preprocessing, feature encoding, fusion strategies, and classification layers, forming the backbone of the model's predictive capability.

Overview of the proposed framework

DermaScanAI is a novel end-to-end, explainable DL Framework accompanied by clinical metadata. The framework encompasses data preprocessing, feature extraction, attention-based feature refinement, metadata fusion, and explainable multi-class classification. Together, each stage — artefact removal, feature discrimination, context integration, and model interpretability — is carefully tailored to address the associated challenges with high clinical relevance in skin cancer detection.

The system takes dermoscopic images as input — with or without patient metadata such as patient age, sex and site anatomy. In a first step, we apply a pre-processing pipeline to the raw images to further clean visual artefacts (e.g., hair strands) and homogenise the lighting. This process improves border visibility and maintains colour uniformity across images, allowing a feature extractor to focus on lesion contours rather than noise artefacts. We achieve this through augmentation techniques for dataset imbalance and better generalisation of the model, as shown in Fig. 1.

Here, the images have been received by the system. At the core of the system is the SkinNet-HDA model, which, after preprocessing, serves as a backbone for feature extraction. We select ConvNeXt-Tiny[48] as the backbone for feature extraction, a convolutional neural network architecture that effectively harnesses local spatial patterns in lesions. A mechanism is used to improve the signal representation. The spatial attention module is on important feature channels and on some key regions of the lesion area. Those refined features are then fed to a lightweight transformer encoder to capture long-range contextual dependencies in the dermoscopic image, and they allow models to capture local details and long-range spatial dependencies.

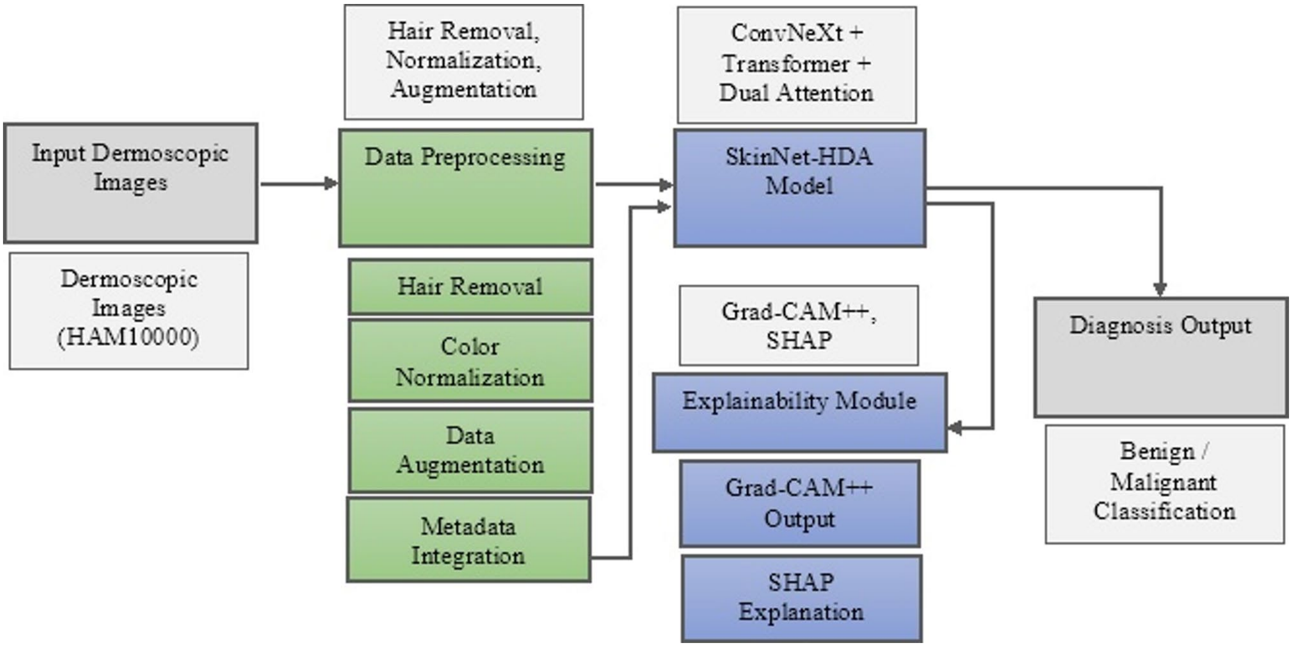

Fig. 1. Overall system architecture of dermaScanAI framework for automated skin lesion classification and clinical integration.

At the same time, the patient metadata from the encoder is merged with the extracted image features via a metadata fusion layer. Clinical background features can provide context for the model-generated visual features and give them interpretation, enabling a more fully formed characterisation of a lesion. Finally, a fully connected classifier is applied to the last feature representation, and the lesion's output category is predicted based on the predefined output classes.

To nurture clinical trust and transparency, the framework was designed with an explainability module. We apply Grad-CAM + + to produce heatmaps showing which areas of the image contribute to the model's decision-making. SHAP analysis is also performed to measure the contribution of image and metadata features to the final classification output. This multi-method explainability plays a critical role in helping the clinician confirm that the model is indeed looking at the right lesion areas and how much the patient-specific factors have played in the prediction.

The proposed DermaScanAI framework yields robust, explainable skin cancer detection by integrating state-of-the-art convolutional and transformer architectures, two attention mechanisms, and unimodal and multimodal data. It can scale and adapt to real-world clinical workflows because it is modular.

Dataset description

We evaluate the proposed DermaScanAI framework on a HAM10000, which is an abbreviation for "Human Against Machine with 10,000 training images"[86] dataset. This is among the most extensive and widely used public datasets for dermoscopic image analysis in skin lesion classification. It includes a variety of high-resolution dermoscopic images from different clinical sources, representing a wide range of lesions, anatomical regions, and patients, thereby reflecting realistic clinical variability.

HAM10000 dataset —contains 10,015 dermoscopic images. Each image is labelled with a diagnosis of the lesion, which was confirmed by histopathological analysis, in vivo confocal microscopy, clinical follow-up, or expert consensus, where applicable. Although the dataset contains seven lesion classes, we are only considering five clinically relevant classes for the proposed classification task, as they provide an adequate number of samples. Criteria include melanoma, melanocytic nevus, benign keratosis-like lesions, basal cell carcinoma, and actinic keratoses. Vascular lesions and dermatofibroma were the other two classes present in the dataset; their limited representation made it challenging to train the deep learning models on them, so they were excluded from this study.

The dataset consists of dermoscopic images with associated clinical metadata, giving clinical context for each patient. The analysed metadata also included patient age, sex, and the lesion's anatomic location: trunk, lower extremities, or face. Adding this metadata during training helps our model consider both clinical risk factors and site-specific lesion prevalence for dermatologic diseases when making predictions.

Once the dataset was cleaned and preprocessed, a stratified data split was performed to balance the class distribution across the training, validation, and test sets, and then proceeded to the model training step. For instance, 70% for the training set, 15% for the validation set, and the remaining 15% for the test set. To organise the data, a directory structure was set up based on class labels and split types, resulting in an intuitive folder hierarchy directly compatible with deep learning data loaders.

The dermoscopic images in JPEG format, at different resolutions, with a standard input size of 224 × 224 pixels, as part of pre-processing. To enable the integration of clinical metadata in the deep learning model, we encoded the clinical metadata fields as numerical vectors. Moreover, the dataset was enriched with picture rotation, horizontal flipping, and image colour jittering, which are examples of data augmentation techniques. These augmentation techniques help the model learn more effectively while also reducing the dataset's inherent class imbalance.

In conclusion, HAM10000 provides a solid and clinically relevant baseline for assessing the performance of the DermaScanAI framework. The data includes numerous lesion types with clinical metadata that closely align with the goals of our study towards developing and validating an Automated skin cancer detection system using an explainable deep learning framework. Table 2 presents key notations and their corresponding definitions used throughout the mathematical formulation of the proposed framework.

Data preprocessing pipeline

Proper preprocessing of dermoscopic images is crucial to enable subsequent deep learning stages to focus on features related to the lesion rather than on misleading visual artefacts. Note that the HAM10000 dataset has several imperfections and limitations, including hair strands, image illumination variation, and irrelevant markings, which can reduce classification accuracy. Hence, a multi-stage preprocessing pipeline is carried out to clean and standardise the data before it is passed to the model for training.

In the morphological operation, the first stage is hair artefact removal. In particular, a black-hat morphological transformation is applied to the grayscale image to emphasise hair-like structures. That operation is mathematically expressed as in Eq. 5.

$$B = (I \cdot K) - I \tag{5}$$

Where I is the grayscale dermoscopic image, K is the structuring element, and denotes the morphological close operation. Blackhat result for B shows dark structures like hair. Then, thresholding on B is performed to obtain a binary mask for hair regions, and one of the inpainting algorithms shown in Fig. 2 is applied to the masked image, with the goal of removing and bounding the masked areas using pixel information from surrounding pixels. This inpainting method restores occluded lesion areas caused by hair without introducing additional visual artefacts.

Notation	Description
$I \in \mathbb{R}^{H \times W \times 3}$	Input dermoscopic image of resolution $H \times W$ with 3 RGB channels
$M \in \mathbb{R}^{k}$	Metadata vector containing patient age, sex, and lesion site
$y \in \{1,2,\ldots,C\}$	Ground truth lesion class label (multi-class classification)
$f(\cdot)$	Classification function parameterized by Θ
Θ	Model parameters to be optimized
$X^{(l)}$	Output feature map of the l-th convolutional layer
$W^{(l)}, b^{(l)}$	Weights and bias of the l -th convolutional layer
F	Intermediate feature map from ConvNeXt backbone
$GAP(\cdot)$	Global average pooling operation
F_{CA}	Feature map after channel attention module
F_{SA}	Feature map after spatial attention module
$\odot$	Element-wise multiplication
$[\cdot;\cdot]$	Concatenation operation along the feature dimension
$\alpha_{k}^{(c)}$	Grad-CAM++ weighting coefficient for the k -th feature map and class cc
A^{k}	Activation map of the k-th channel in the final conv layer
$L_{Grad-CAM++}^{(c)}$	Grad-CAM++ heatmap for class c
φ_{i}	SHAP value (contribution) of feature i
$\mathcal{L}_{cls}$	Classification loss (focal loss in this work)
λ	Weight decay factor in AdamW optimizer
η_{t}	Learning rate at iteration t
T_{max}	Maximum number of iterations for cosine annealing scheduler
Z	Fused feature vector combining image and metadata features
$\hat{y}$	Model output prediction vector

Table 2. Notations and descriptions used in the mathematical formulation of the proposed SkinNet-HDA framework.

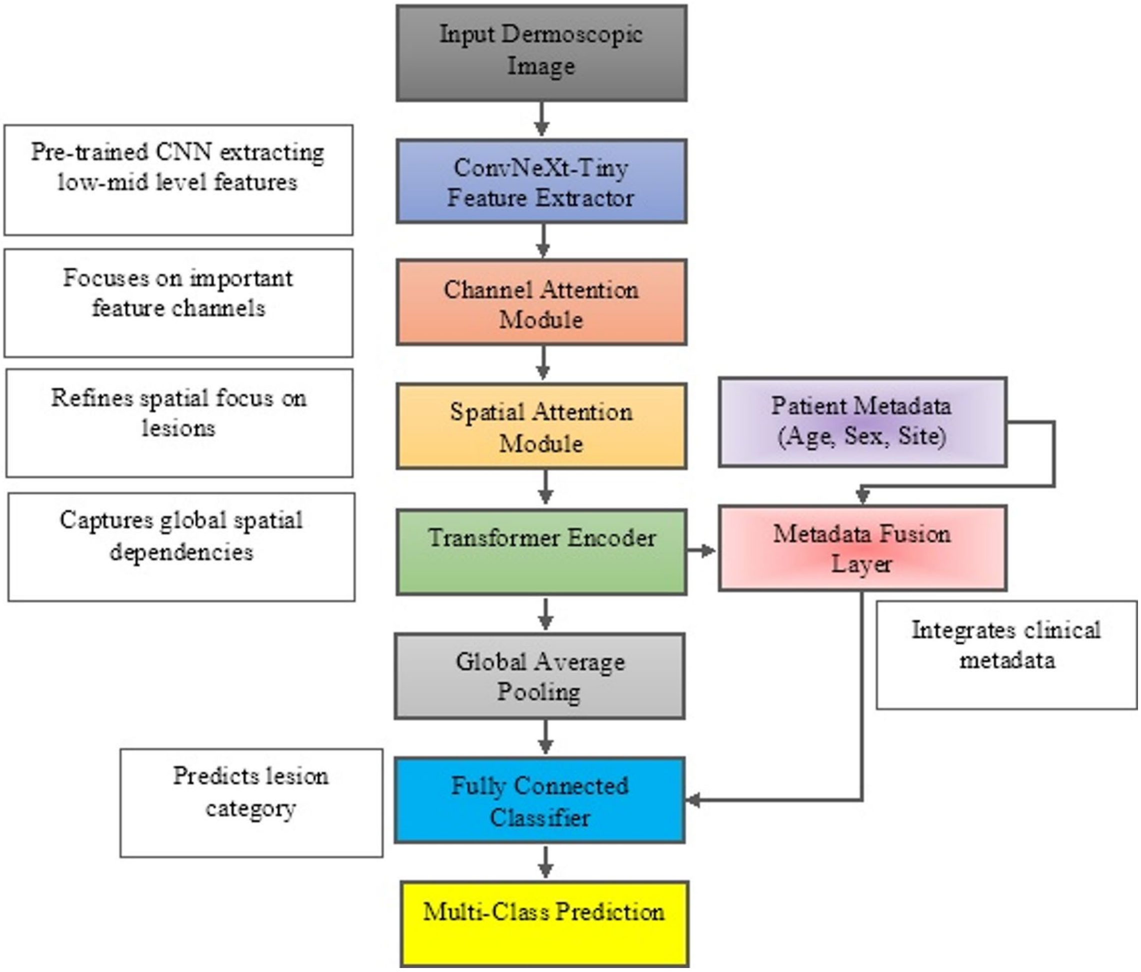

Fig. 2. Detailed model architecture of SkinNet-HDA incorporating hybrid dual attention and multi-level feature extraction for skin lesion classification.

Normalising colour to reduce illumination variations and colour changes caused by image capture with different devices is mandatory for hair removal and cleaning. The Shades-of-Grey colour constancy algorithm, which scales each colour channel by functions of reflectance statistics. The color channels I_{norm} are normalized as in Eq. 6.

$$I_{norm}^{(c)} = \frac{I^{(c)} \cdot \mu}{\left(\sum_{c}\left(\frac{1}{N}\sum_{i=1}^{N}\left(I_i^{(c)}\right)^p\right)^{\frac{1}{p}}\right)} \tag{6}$$

Where $I^{(c)}$ is the input color channel c, μ is a normalization constant, there are a total of N pixels, and p is the Minkowski norm (usually $p = 6$). To normalise pixel values across all channels so they are evenly distributed, making lesions more easily visualised and maximising robustness in feature extraction.

Numerical vectorisation of patient clinical metadata (age, sex, lesion site) for integration with the image features. Min-Max scaling is applied to the age values. Sex (binary categorical variable): 0 for male; 1 for female Each anatomical site category are mapped to unique integer identifiers, delimited in a metadata vector denoted as $M \in \mathbb{R}^k$. The full metadata vector is specified as in Eq. 7.

$$M = [Age_{norm}, Sex_{bin}, Site_{code}] \tag{7}$$

where each entry represents a normalized or encoded metadata field.

A combination of data augmentation techniques is then performed to tackle the problem of class imbalance, and to enhance model generalization. Such as random rotations to imitate image orientation, horizontal flip for spatial diversity, and color jittering to change brightness and contrast. By using this augmentation, we generate more diverse training data facilitating the model to learn a more robust representation on different conditions of the images.

Lastly, the sizes of all images are set to 224×224 pixels to match the input size of the SkinNet-HDA model. This forces computational efficiency and is compatible with pre-trained CNN architectures.

We highlight that this data preprocessing pipeline is crucial for enhancing the quality and standardisation of dermoscopic images whilst embedding appropriate clinical metadata. We then perform artefact removal, colour normalisation, metadata encoding, and data augmentation, all to shape the data into a clean, explainable space optimal for skin cancer classification.

Proposed SkinNet-HDA model architecture

In this article, we propose a hybrid DL architecture, SkinNet-HDA, for extracting robust, explainable feature representations of dermoscopic images while enabling the incorporation of patient-specific metadata. The proposed model fuses a convolutional backbone into the architecture, incorporates bilinear feature attentions, employs a transformer encoder and adopts a metadata fusion approach to accomplish multi-class skin lesion classification.

In the first stage of the model, we use ConvNeXt-Tiny, specialising in visual feature extraction. ConvNeXt still adopts the conventional convolutional layer introduced in (1). This backbone can extract multi-scale hierarchical features (local texture patterns, colour variations, and lesion boundary details) from the pre-processed thermoscope images.

We present a dual-attention mechanism with channel attention and spatial attention modules to improve feature selectivity. The channel attention module focuses on the most semantically meaningful feature channels. This is accomplished using an SE block, in which a set of feature maps is globally pooled and processed by the network. The recalibrated feature maps F_{CA} are calculated as in Eq. 8.

$$F_{CA} = F \odot \sigma \left(W_2 \delta \left(W_1 GAP\left(F\right)\right)\right) \tag{8}$$

where F are the input feature map from ConvNeXt, GAP(·) denotes global average pooling, W_1 and W_2 are learnable weight matrices of the fully connected layers, δ(·) is the ReLU activation, σ(·) is the sigmoid activation, and ⊙ denotes channel-wise multiplication.

Then, the spatial attention module refines the feature map by emphasizing on the most informative spatial regions after channel attention. The spatial attention map is calculated through a concatenation of the average-pooled and max-pooled features along the channel dimension and passing them through a convolution and a sigmoid activation. This gives rise to the refined feature maps F_{SA} as in Eq. 9.

$$F_{SA} = F_{SA} \odot \sigma \left(Conv\left(\left[AvgPool\left(F_{CA}\right); MaxPool\left(F_{CA}\right)\right]\right)\right) \tag{9}$$

Where $[\cdot\,;\cdot\,]$ is time-wise concatenation and $Conv(\cdot\,)$ is a convolutional layer.

Then, captured long-range spatial dependencies from the lesion regions via the output of the dual attention mechanism (as in Eq. 2), with layer normalization, and feed-forward transformations. Integrating contextual relationships between distant image patches helps the model to obtain a holistic understanding of the whole lesion structure.

To add clinical context, we concatenate M, our image-agnostic metadata vector previously defined in Eq. 7, with the feature vector of the image extracted after the transformer encoder. Collectively, F_{img} is the pooled image feature representation. Combined of all the features vector, Z will be the vector we will have Eq. 10.

$$Z = [F_{img}; M] \tag{10}$$

Where $[\cdot\,;\cdot\,]$ denotes vector concatenation. Through this fusion, the classifier can make predictions $\widehat{y}$ based on both visual and clinical clues simultaneously.

A fully connected classifier then takes the vector Z as input and uses a softmax written as below, to provide a mapping from the combined feature space to a probability distribution over the C lesion classes. Predicting output at final stage is done via softmax layer as in Eq. 11.

$$\widehat{y} = softmax\left(W_{fc}Z + b_{fc}\right) \tag{11}$$

where W_{fc} and b_{fc} denote the weight matrix and bias term of the fully connected classifier, respectively.

The entire skinNetHDA architecture is optimised from end to end with respect to the focal loss function described in Eq. 1, allowing the model to learn context-aware, discriminative, and interpretable representations for skin lesion classification with great accuracy.

Loss function and optimization

The HAM10000 dataset is highly imbalanced by default, with some lesion categories represented by very few samples. To tackle this issue, the SkinNet-HDA model uses the focus loss function, as previously defined in Eq. 1. Instead of a fixed weighting of samples, the focal loss dynamically adjusts the cross-entropy loss by a factor that reduces the loss for a small number of hard, minority-class examples. This is used to reduce the model's overfitting when handling majority classes and to increase its sensitivity to malignant lesions in melanoma, etc.

Denote p_t the probability of the ground-truth class we predicted, and γ the focusing parameter.

$$\mathcal{L}_{focal} = -\alpha_{\,t}(1 - p_t)^{\gamma}\, log\left(p_t\right) \tag{12}$$

Here, α_t is a weighting factor to adjust the weight of a specific class. In the scope of the current work, γ is empirically fixed as 2.0 to up-weight easy examples and α_t is calculated according to the reciprocal class frequency distribution. This focal loss for a mini-batch of N samples is given by Eq. 12.

$$\mathcal{L}_{cls} = \frac{1}{N}\sum{}_{i=1}^{N}\mathcal{L}_{focal}^{(i)} \tag{13}$$

Then, this batch-level loss $\mathcal{L}_{cls}$ corresponds to the optimization objective defined above in Eq. 13. The AdamW Optimizer is used for model optimization because it has the feature of preventing overfitting and improving generalization. This formulation gives better control over the model regularisation since it decouples the weight decay term from the gradient update unlike the standard Adam optimiser. The update step, for the model parameters Θ at iteration is given in Eq. 14.

$$\Theta^{(t+1)} = \Theta^{(t)} - \eta_t \cdot \widehat{g}^{(t)} - \lambda\,\Theta^{(t)} \tag{14}$$

Where η_t is learning rate on iteration t, $\widehat{g}^{(t)}$ is the biased corrected gradient and λ is weight decay factor. During training, we additionally apply a Cosine Annealing learning rate scheduler for better convergence. The learning rate η_t at iteration t follows the cosine decay as expressed in Eq. 15.

$$\eta_t = \eta_{min} + \frac{1}{2}\left(\eta_{max} - \eta_{min}\right)\left(1 + cos\left(\frac{\pi\,t}{T_{max}}\right)\right) \tag{15}$$

where, η_{min} and η_{max} are the minimum and maximum learning rates respectively, and T_{max} denotes the max number of iterations. The benefit of this scheduling method is that it allows the learning rate to decay smoothly, encouraging stable convergence and preventing abrupt drops.

A sequence is used to set up the model parameters in a standard way (for example, Xavier initialisation for linear layers). Model training is carried out over several epochs, with early stopping based on validation loss and accuracy. This helps the final model achieve the best possible generalisation without overfitting to the training data.

To conclude, the focal loss for class imbalance, the AdamW optimiser for a smooth convergence path, and scheduling for an optimal learning mechanism all contribute to the proposed training strategy for the SkinNet-HDA model, aiming to achieve robust skin cancer classification.

Explainability module

Especially in a clinical setting, it is essential to build trust within the network of healthcare professionals and ensure that decisions are based on clinically relevant evidence. DermaScanAI framework with an explainability module that fuses gradient-based and feature attribution methods to provide visual and feature-level explanations of model predictions. This module improves clinical transfer validation by showing to what extent the predicted findings relate to specific lesions or lesion-relevant metadata that ultimately determine the outcome.

One of the explainability methods used in this paper is Grad-weighted Class Activation Mapping Plus Plus 1, which produces a heatmap showing the important regions of the dermoscopic image for classification. Where the original Grad-CAM uses first-order gradients, Grad-CAM + + supplements them with higher-order derivatives that improve the localisation of the relevant area, especially when there are many contributing areas in the image. Given a specific class c that we want to compute the Grad-CAM + + heatmap $L_{Grad-CAM++}^{(c)}$ is expressed as in Eq. 16.

$$L_{Grad-CAM++}^{(c)} = ReLU\left(\sum{}_k \alpha_k^{(c)} A^k\right) \tag{16}$$

Where A^k is the k-th feature map from the last convolutional block and $\alpha_k^{(c)}$ are the weighting coefficients defined from the gradients of the target class score feature maps. With ReLU activation, only positively influencing regions will be visualised on the heatmap. The heatmaps are superimposed on the original dermoscopic images to illustrate which areas of the lesion contribute to the model's decision (allowing clinicians to validate that the areas the model is focusing on correspond to clinically relevant patterns of lesions).

Besides improving visual interpretability, we further integrate Shapley Additive exPlanations (SHAP) into our framework to quantify the contributions of clinical metadata and image-derived features to the final prediction. SHAP values provide a feature attribution score for each input variable, allowing identification of whether each variable positively or negatively affects the input, e.g., the predicted lesion class. The SHAP explanation ensures that this additivity property holds for each input instance, where the sum of contributions from individual features approximates the difference between the model output and the expected output across the entire dataset. More formally, the SHAP value for a feature is defined by Eq. 17.

$$\varphi_i = \sum{}_{S\subseteq F\setminus\{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!}\left[f(S\cup\{i\}) - f(S)\right] \tag{17}$$

with F being the set of all features, S being a set of features excluding i, and $f(S)$ being the output of the model given the subset SS. This holds ensures that the model output is fairly distributed across different features.

The explainability module includes Grad-CAM + + heatmaps at the visual feature level and SHAP explanations at the metadata level, allowing a combined view of how lesion appearance and clinical context factors into the model's decision-making. In addition, if the ground truth lesion masks are available, we will also perform lesion-mask consistency analysis by checking whether the Grad-CAM + + heatmaps indeed detect the lesion regions indicated by the human expert.

The explainability module embedded in SkinNet-HDA can facilitate us in using SkinNet-HDA as a transparent, clinically credible diagnostic tool rather than a black-box classifier. This kind of transparency is vital to driving clinical adoption and helping dermatologists appreciate and believe in the rationale behind automated diagnostics.

Algorithmic implementation

In this section, we describe the sequence of steps implemented by the LungCare-AI framework, converting the architectural design into a computational pipeline. It streamlines the way training proceeds, from data selection to feature encoding via convolutional and attention layers to the final output. We also provide the pseudocode for the integrated workflow to make the proposed deep learning (DL) approach clearer and more reproducible.

In Algorithm 1, we describe the entire preprocessing and augmentation pipeline, which preprocesses dermoscopic images and their associated clinical metadata, preparing them for input to the SkinNet-HDA model. The pipeline starts with a raw dermoscopic image containing multiple artefacts, including hair strands, non-uniform illumination, and background noise. To address these issues, the image is initially converted to grayscale and then subjected to a morphological blackhat operation to identify hair-like structures. Thus, it increases the visibility of linear structures on the background, providing a binary mask of probable hairy regions. The binary mask is thresholded, and the detected hair regions are inpainted, meaning the algorithm fills in missing content by interpolating the values of surrounding pixels. The lesion's boundaries are so preserved without distortion.

After removing the artefacts, the pipeline normalises the image using the Shades-of-Grey (SOG) algorithm. This method performs illumination compensation by balancing reflectance offset and lighting variations, and colour bias across devices. The same is true for lesion appearance; however, if all samples are normalised and have similar acquisition conditions, the lesion can appear the same across all samples. After which, we resize it to 224 × 224 pixels to match the model's input shape.

A sequence of augmentation operations is performed on the preprocessed image. These are random rotations, Horizontal Flips, and Color Jitter. In this way, every augmented image is a plausible version of the original lesion, encouraging the model to learn invariant representations of lesion features.

Simultaneously, we preprocess the clinical metadata linked to each image. The patient's age is normalised to values between 0 and 1; the sex attribute is modelled as a binary variable; and the lesion's anatomical site is mapped to an integer-valued code. The fields are concatenated to generate a numerical metadata vector that can be effectively integrated with the model's image-dependent features.

The preprocessed image, the augmented image, and the metadata are then passed as input to the SkinNet-HDA model, which extracts and classifies features (the output of Algorithm 1). This preprocessing pipeline yields uniform, artefact-free data enriched by context, enabling robust and explainable learning in the following steps:

Algorithm 2 focuses on the preprocessing functions and illustrates the workflow of the SkinNet-HDA model, which takes preprocessed dermoscopic images and encoded clinical metadata as inputs and produces a multi-class lesion classification as output. As shown in the workflow, it starts with the augmented dermoscopic image, which is fed through the ConvNeXt-Tiny backbone. At this stage, hierarchical image features are extracted, comprising both low-level patterns (e.g., edges of pathology) and high-level semantic features of pathology morphology. These convolution operations adhere to the previous formulation, yielding an initial representation of the region of interest containing the lesion.

A dual-attention mechanism is then applied to refine the extracted features and emphasise informative features. The feature maps are first improved using the channel attention module, which emphasises particularly relevant channels for lesion characterisation. This is done by taking a two-layer fully connected network and

Input: Raw dermoscopic image I, metadata fields (age, sex, site)
Output: Preprocessed image I_{aug}, encoded metadata vector M

1. Convert I to grayscale and apply morphological blackhat operation to detect hair artifacts B using Equation (5).
2. Threshold B to obtain binary hair mask and apply inpainting to restore occluded regions.
3. Apply Shades-of-Graycolor normalization on I to produce illumination-corrected image I_{norm} as per Equation (6).
4. Resize I_{norm} to 224×224 pixels.
5. Apply random rotation, horizontal flipping, and color jittering to obtain augmented image I_{aug}.
6. Normalize patient age to [0,1] range, encode sex as binary (0 or 1), and map lesion site to integer code.
7. Concatenate encoded metadata to form vector M as defined in Equation (7).
8. Return I_{aug} and M.

Algorithm 1. Data preprocessing and augmentation pipeline.

Input: Preprocessed image I_{aug} , encoded metadata vector M
Output: Predicted lesion class y

1. Pass I_{aug} through the ConvNeXt-Tiny backbone to extract feature maps F as per Equation (1).
2. Apply channel attention on F to obtain F_{CA} using Equation (8).
3. Apply spatial attention on F_{CA} to obtain F_{SA} using Equation (9).
4. Pass F_{SA} through the transformer encoder to capture long-range dependencies using Equation (2).
5. Apply global average pooling to flatten the transformer output into vector F_{img} .
6. Concatenate F_{img} and metadata vector M to form the fused feature vector Z as defined in Equation (10).
7. Pass Z through a fully connected classifier and apply softmax to compute prediction $\hat{y}$ as per Equation (11).
8. Return $\hat{y}$.

Algorithm 2. SkinNet-HDA model workflow.

Input: Training dataset $D = \{(I_i, M_i, y_i)\}_{i=1}^{N}$, model parameters Θ
Output: Optimized model parameters Θ^*

1. Initialize Θ, AdamW optimizer, and Cosine Annealing learning rate scheduler as per Equations (13) and (14).
2. For each training epoch:
   3. For each mini-batch $\{(I_{aug}, M, y)\}$:
   4. Forward pass: Compute predicted class $\hat{y}$ using Algorithm 2.
   5. Compute focal loss $\mathcal{L}_{focal}$ between $\hat{y}$ and y as defined in Equation (1).
   6. Backpropagate gradients and update Θ using AdamW optimizer.
   7. Update learning rate using cosine annealing schedule.
   8. Evaluate model performance on validation set.
   9. Save checkpoint if validation AUC improves.
3. Return optimized model parameters Θ^*.

Algorithm 3. Training procedure for SkinNet-HDA with focal loss.

a sigmoid activation, formulated in Eq. (8). The feature maps obtained include channels that have been given greater importance because they are clinically relevant.

The spatial attention module retrieves the most informative spatial representations from the set of feature maps corresponding to the dermoscopic image after channel refinement, thereby improving feature representation performance in critical spatial regions. This attention module extracts attention maps from spatial features pooled by average and max pooling, then passes them through a convolutional transformation and a sigmoid activation, as per Eq. 9. The generated feature maps are retained at the spatial locations that contribute most to lesion classification.

Attention-Crossed Features is a lightweight transformer encoder that models the global contextual relationships between different regions of the lesion. A key property of the transformer encoder is that it applies self-attention mechanisms (Eq. 2), allowing the model to attend to image patches that may be far apart in the input image but are jointly responsible for the classification prediction. Global average pooling of the output yields a compact image feature vector.

Simultaneously, the clinical metadata vector, based on the patient's normalised age, one-hot-encoded sex, and lesion site, is prepared for concatenation. The combined feature vector is obtained by concatenating the image feature vector and the metadata vector, as shown in (10). By integrating these modalities, the model can account for visual features within the context of patient-specific clinical information, thereby improving prediction contextuality.

Finally, we feed the fused feature vector into a fully connected classification head, consisting of a softmax activation to compute the class probabilities, as formalized in Eq. (11). The output is the softmax probability distribution over lesion classes, enabling the model to classify the input lesion as melanoma, nevus, or neither.

The SkinNet-HDA workflow is a single end-to-end architecture that integrates convolutional feature extraction and attention-guided refinement, followed by transformer-based global context modeling and metadata fusion. Utilizing a model that integrates image-derived features and clinical context features for robust and explainable skin cancer classification.

In Algorithm 3, we depict the supervised training process to carefully optimize SkinNet-HDA parameters to achieve robustness and balance in objective-oriented skin lesion classification. The input to the model of data set preprocessed dermoscopic images, encoded metadata vectors, and their respective ground-truth lesion class labels. We seek to update the model parameters Θ iteratively such that our model learns to classify all lesion categories, despite the natural imbalance in the dataset.

The model parameters, cap theta, are first set up, and the AdamW parameter optimizer and Cosine Annealing learning rate scheduler are also set up. We use AdamW due to its decoupling of weight decay from gradient-based updates, which has been shown to improve the regularization and stability of convergence. The Cosine Annealing scheduler gradually reduces the learning rate in a cosine manner as training proceeds, which allows the model to slowly switch from the exploration phase to the fine-tuning phase according to Eq. (14).

Within each epoch, the process is done on mini-batches from the training dataset to take advantage of parallelization and stable estimation of the gradients. The processed images and metadata vectors for each mini-batch iteration are fed into the SkinNet-HDA model, which implements the steps outlined in Algorithm 2. For each input sample, the model outputs a predicted class probability distribution of classes.

The foreground loss can be calculated with the focal distribution of classes and the ground-truth label, as expressed in Eq. (1). Focal loss alleviates class imbalance by down-weighting the contribution of easily classified examples and concentrating the learning on hard minority class instances. The mean focal loss of mini-batch is calculated so that we build the batch-level loss $\mathcal{L}_{cls}$ according to Eq. (12).

Backpropagation is used to calculate the loss gradients Θ. The AdamW optimizer changes the model parameters using these gradients, performing weight decay regularization as shown in Eq. (13). The learning rate is decreased after each epoch by the cosine annealing schedule, reducing the step size as we approach convergence.

During training, the classification accuracy and AUC are monitored on the validation set. The best-performing model is the one whose model checkpoint corresponds with the highest validation AUC. The validation accuracy patterns are monitored and unnecessary training epochs.

This training procedure iteratively optimizes SkinNet-HDA model parameters across several epochs, yielding an optimal parameter set Θ^* that maximizes classification accuracy while balancing performance over all lesion classes. This process will ensure the model's good deduction for unseen dermoscopic images, and will also handle the class imbalance issue well with the help of focal loss.

Algorithm 4 applies the trained SkinNet-HDA model to the test images and obtains the prediction results, from which we can generate visual- and feature-level explanations as outlined in Algorithm 4. This explainability pipeline increases the explainability of deep learning systems and enables clinicians to understand why the system made its classification, allowing them to evaluate the trustworthiness of the deep learning output.

Let us start with a step-by-step overtake of implementation as it starts with the fed self-processed dermoscopic test image and its classified clinical metadata vector. Based on these inputs, we calculate the class probability distribution $\widehat{y}$, providing them to the trained SkinNet-HDA model (parameterized by Θ^*) after a forward pass. Prediction of the model includes the class c, which is the lesion class having the most probability for for which the explanations are been generated.

The convolutional feature maps of the SkinNet-HDA model need to be visual interpretable, so the GradCAM++ method is used for this. The first step involves calculating the gradients of the target class score $\widehat{y}_c$ w.r.t each feature maps A^k by backpropagation. These gradients capture how the change in each feature map results in a change in the predicted score for the class c, and Grad-CAM++ algorithm uses these gradients to compute the weighting coefficients $\alpha_k^{(c)}$ that indicate the class prediction. These coefficients are then used to compute a yield the final heatmap $L_{Grad-CAM++}^{(c)}$, formulated as in Eq. (15). The resulting heatmap is a into a ReLU activation which highlights regions that is then superimposed on the dermoscopic image to depict the area of dermoscopic image.

At the same time, we do feature-level interpretability with SHapley Additive exPlanations (SHAP). Our entire feature set includes the flattened image feature vector along with the encoded metadata as input features. It breaks down the output of the model by computing the marginal feature i to the final class prediction sure that

Input: Trained SkinNet-HDA model Θ^*, test image I_{aug}, metadata vector M, target class c

Output: Grad-CAM++ heatmap $L_{Grad-CAM++}^{(c)}$, SHAP feature contributions $\{\phi_i\}$

1. Perform a forward pass on I_{aug} and M to compute class predictions $\hat{y}$.
2. For Grad-CAM++ explainability:
   3. Compute gradients of the class score $\hat{y}_c$ with respect to feature maps A^k.
   4. Calculate Grad-CAM++ weights $\alpha_k^{(c)}$ using higher-order gradients.
   5. Generate heatmap $L_{Grad-CAM++}^{(c)}$ as defined in Equation (15).
   6. Overlay $L_{Grad-CAM++}^{(c)}$ on I_{aug} to visualize lesion focus.
3. For SHAP explainability:
   7. Flatten image features and concatenate with metadata vector M to form the input feature set F.
   8. Compute SHAP values ϕ_i for each feature i using Equation (16).
   9. Interpret feature contributions for clinical insight.
4. Return $L_{Grad-CAM++}^{(c)}$ and $\{\phi_i\}$.

Algorithm 4. Explainability generation using Grad-CAM++ and SHAP.

all contributions sum up to the model output minus the expected output. SHAP values φ_i are calculated based on the Eq. (16), building on the concepts from cooperative game theory, that fairly distributes the prediction output into the features that cooperated in the prediction. The output is a list of attribution scores that indicate the positive or negative influence that each input feature has the predicted class.

The outputs of our explainability module, which are performed as post-hoc explainability on the original task results. The final outputs include a Grad-CAM + + heatmap that provides visualization to highlight regions of the lesion contributing to the decision, and SHAP values that explain the contribution of visual and metadata features for data classification. These outputs jointly provide a detailed rationale for the model's decision, thereby fostering clinical trust and ensuring the interpretability required in real-world diagnostic systems.

Experimental setup

The experimental design of the proposed DermaScanAI framework to comprehensively evaluate the performance of the SkinNet-HDA model on the HAM10000 dataset under controlled evaluation conditions is presented. All experiments were conducted on a workstation with 64GB RAM and an NVIDIA RTX 3090 GPU—implementation was done using PyTorch 2.0 and Ubuntu 20.04. The training and evaluation pipelines incorporated libraries for TorchMetrics, torchvision, OpenCV, and PyTorch Grad-CAM++.

We preprocessed the HAM10000 dataset as described in Sect. 4.3 and stratified it into train/valid/test splits for each of the seven classes (70%, 15%, and 15%, respectively). Class distributions across splits were preserved as much as possible to create a balanced evaluation. During training, images were resized to 224 × 224 pixels and metadata fields (age, sex, and lesion site) were also converted to numerical vectors. To generalize better and address class imbalance.

Training of the SkinNet-HDA model was performed for 30 epochs with a batch size of 32. Importantly, the initial learning rate was set to 1×10^{-4}, and ooptimised through the AdamW optimiserin accordance with Eq. 13. During training, a cosine annealing scheduler [loshchilov2016sgdr] was used to adapt a minimum learning rate of 1×10^{-6}. The decay pattern is defined according to the following Eq. (14). We used focal loss as the main classification loss to mitigate class imbalance (see Eq. 1).

All hyperparameter tuning was carried out with a 10% holdout of the training set, referred to as the validation set. Model performance was tracked on the validation set after every epoch, and the model checkpoint with the best validation AUC was then evaluated on the test set. To avoid overfitting, we used five epochs on validation accuracy as feedback.

We maintained the overall metrics per class, and macro-averaged and weighted AUC scores as our evaluation metrics. To qualitatively assess model attention and feature contributions, Grad-CAM + + heatmaps and SHAP feature attribution values were computed on a subset of correctly and incorrectly classified test samples (explainability evaluation).

For reproducibility, the same random seed (42) is used for data shuffling, model initialisation, and split generation. To mitigate variability from random initialisation, the performance numbers are reported as means across three independent repeats of all experiments. All in all, the proposed experimental setup provides a fair, thus reproducible, laboratory setting, confirming the efficiency of the proposed skinnet-HDA hybrid model for dermoscopic images, with results that can also be easily explained.

Experimental results

The experimental results section thoroughly evaluates the proposed framework for detecting skin cancers using the HAM10000 dataset. It describes the experiment setup, performance measures, and how it compares to other models. It also showcases ablation studies and explainability visualisations to demonstrate the model's robustness, accuracy, and interpretability when performing clinical skin lesion classification tasks.

Dataset description

HAM10000 dataset[85]: One of the most referenced benchmarks for automated skin lesions analysis and classification. The D-Stream dataset includes 10,015 dermatoscopic images collected from diverse populations and sites, representing various pigmentation and lesion types. The dataset consists of seven diagnostic types. The balanced class distribution covers the types of studies typically seen in the clinic and can therefore serve as a reference for developing and evaluating the performance of any deep learning method applied to the dermatological setting.

All images in the dataset are rescaled to 600 × 450 pixels, with all dermoscopic media acquisition methods consistently applied. Lastly, but not least, the photos are labelled and verified by expert dermatologists, which ensures high-quality labels and clinical relevance. We provide the dataset, freely available via the Harvard Dataverse, in a format that promotes reproducibility and benchmarking across all machine learning and medical image analysis studies. In this study, classes are balanced across the splits. Having such a setup is crucial to ensure the reliability of the diagnosis and the generalizability of the proposed framework, particularly for the dominant classes.

Experimental setup

The experimental design was intended to test the ability of the proposed skin cancer classification framework to perform and generalise under highly realistic, reproducible experimental conditions, providing sufficient computational resources to train and evaluate deep learning models. We ran our software computations on Python 3.10 and TensorFlow 2. X and Keras serve as the core deep learning back-end, supported by basic libraries such as NumPy, OpenCV, scikit-learn, and Matplotlib, for data management and representation.

HAM10000 dataset: Images were normalised and resized to 224 × 224 pixels before training, with pixel values scaled to [0, 1]. To improve model generalisation and reduce overfitting, data augmentation was used, including horizontal and vertical flipping, random zooming, rotation, and brightness adjustment. Model training with a batch size of 32, a learning rate of 0.0001, and early stopping determined by the validation loss (to cut out unnecessary epochs).

We reported model performance using standard classification metrics, such as AUC. To ensure the most reliable and unbiased performance estimation, stratified k-fold cross-validation ($k = 5$) was also employed. Each experiment was repeated three times, and results were averaged across random initialisations. Such a strict setup is essential for producing accurate experimental results that genuinely reflect the performance capacity of the multiclass skin lesion classification system.

Evaluation metrics

Given the proposed skin cancer classification system, multiple metrics have been used to evaluate its predictive accuracy. This includes metrics (AUC–ROC) and Globalmetric that indicate accuracy across all classes — Accuracy. However, since the dataset is imbalanced, we have used other metrics to give a balanced representation of the performance.

Precision is defined as the proportion of actual positives among all expected positives; it measures how well the model avoids false positives. Recall, also known as sensitivity, is the number of true positives the model correctly identified, indicating how well it can spot authentic skin cancer. The F1-Score is the Harmonic Mean of precision and recall, which helps balance the two and works very well, especially with imbalanced datasets.

Also, the AUC-ROC (area under the ROC curve) was used to assess the classifier's ability to discriminate between classes at different threshold values. It combines sensitivity and specificity and is thus a more useful measure than accuracy in most multiclass medical classification problems, where both are essential. The combination of these metrics will guarantee that the proposed deep learning framework is clinically meaningful and assessable.

Exploratory data analysis

Exploratory data analysis (EDA) is a critical step before model building, as it provides insights into patterns, distributions, or imbalances, as was the case with the data we used to classify skin cancers. It helps pinpoint biases, outliers, and other characteristics that influence the model's functioning. In this section, visual and numerical explorations are presented to examine the distributions of numeric and categorical features, which help inform model-building and preprocessing decisions.

A big class imbalance can be observed in the HAM10000 dataset where in the majority of skin lesions are belonging to the nevus class as illustrated in Fig. 3. Here, the minority class is substantially outnumbered by the majority class leading to a biased model training. This well-balanced required to achieve stable and fair performance for each lesion class.

This example from the dataset has a balanced distribution, with both classes (0 and 1) equally represented. It is validated that the class imbalance problem in the original dataset is shown twice in Fig. 4. This artificial resampling ensures the DL model is trained adequately with equal numbers of samples from both classes, reducing classification bias and boosting overall performance.

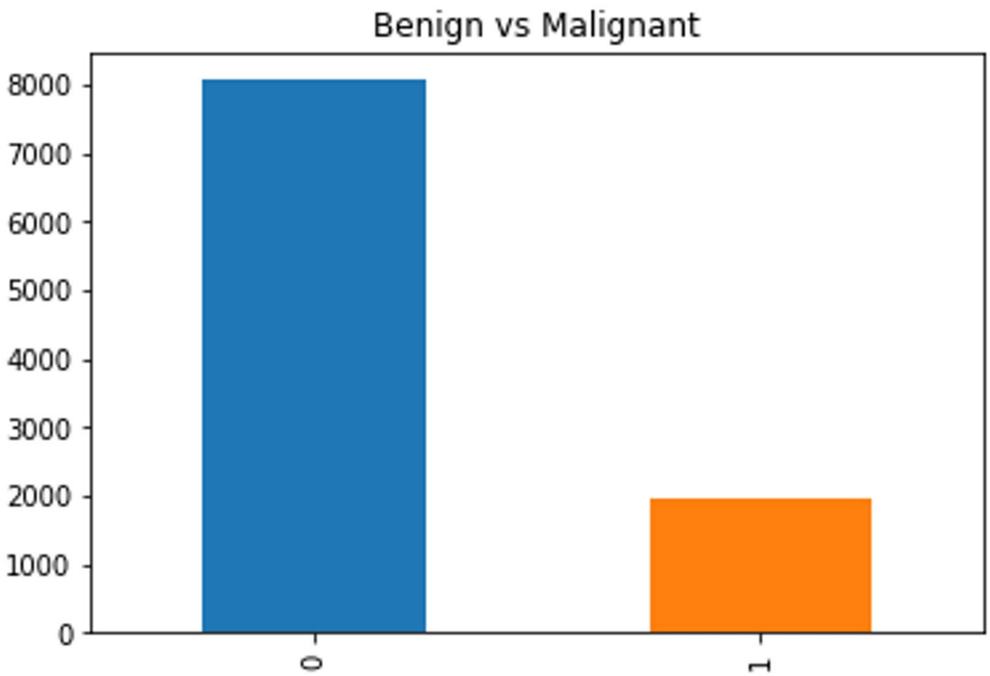

Fig. 3. Distribution of skin lesion classes in the dataset showing class imbalance between majority and minority categories.

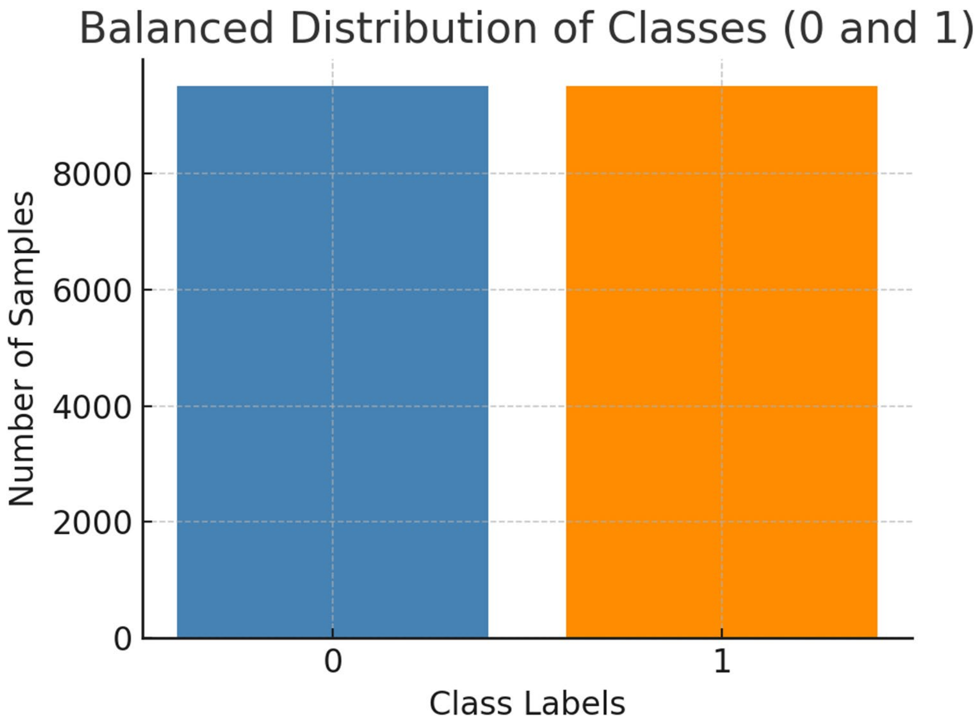

Fig. 4. Balanced distribution of binary classes (0 and 1).

Arrangement of skin lesion images by organ groupings, in descending order of datasets, in Fig. 5. Lesions most frequently occur on the back and lower extremities, followed by the upper limbs and chest. Having this wide range of anatomical sites provides a diversity of lesion contexts and variations, which are necessary for improving classification accuracy in a model.

Comparison of the two types of skin lesion images based on the type of treatment received, as shown in Fig. 6. Most cases were excisions, and the second most common was no treatment. Cryotherapy was used in a smaller share of recipients, while other therapies were given to very few. This distribution provides an overview of the types of clinical interventions performed across different lesion categories in the dataset.

Figure 7 presents representative dermoscopic images from the HAM10000 dataset, illustrating different skin lesion types such as melanoma, nevi, and keratosis. The images exhibit substantial variation in texture, colour, shape, and lesion boundaries, reflecting the high visual complexity of dermoscopic data. This diversity poses significant challenges for automated skin cancer detection and underscores the need for robust and sophisticated deep learning models.

This is shown in Fig. 8, which is a pairplot of average colour intensities (Red, Green, Blue, and Grey) for different skin lesion classes. This juxtaposition is translated into scattered distributions and corresponding histograms, with overarching characteristic clustering profiles for Melanocytic nevi and Actinic keratoses, assisting in the separation of features. These visual insights aid in understanding the role that specific colour features play in differentiating lesion types during classification.

Figure 9 presents representative HAM10000 samples selected to illustrate the high intra-class and inter-class variability in lesion colour, texture, shape, and boundary characteristics. Such variation poses a significant challenge for automated classification systems, which require robust feature extraction and discriminative learning mechanisms to accurately distinguish malignant from non-malignant skin lesions during training and testing.

Figure 10 presents representative test samples comprising a wide variety of lesions with differences in shape, size, pigmentation, and boundary clarity. These images reflect realistic clinical variability and therefore present a challenging test scenario for automated classification. Evaluating the model on such samples helps assess its robustness, generalisation capability, and reliability for real-world diagnostic applications.

Quantitative results

Standard classification metrics and the AUC were then employed to quantitatively evaluate the performance of the proposed MetaBreastNet model in this study. The evaluation dataset consisted of the held-out test set, and analysis was subsequently performed on a per-class basis to assess generalizability across different lesion types—to serve as an external validation.

The performance metrics are reported in Table 3, which shows the proposed model's performance across the test dataset's classes. The outcomes indicate that the model achieves a good trade-off between precision and recall, with reasonably high results for both the benign and malignant classes, signifying that the model can discriminate lesions and that the complete classification framework is reliable and robust.

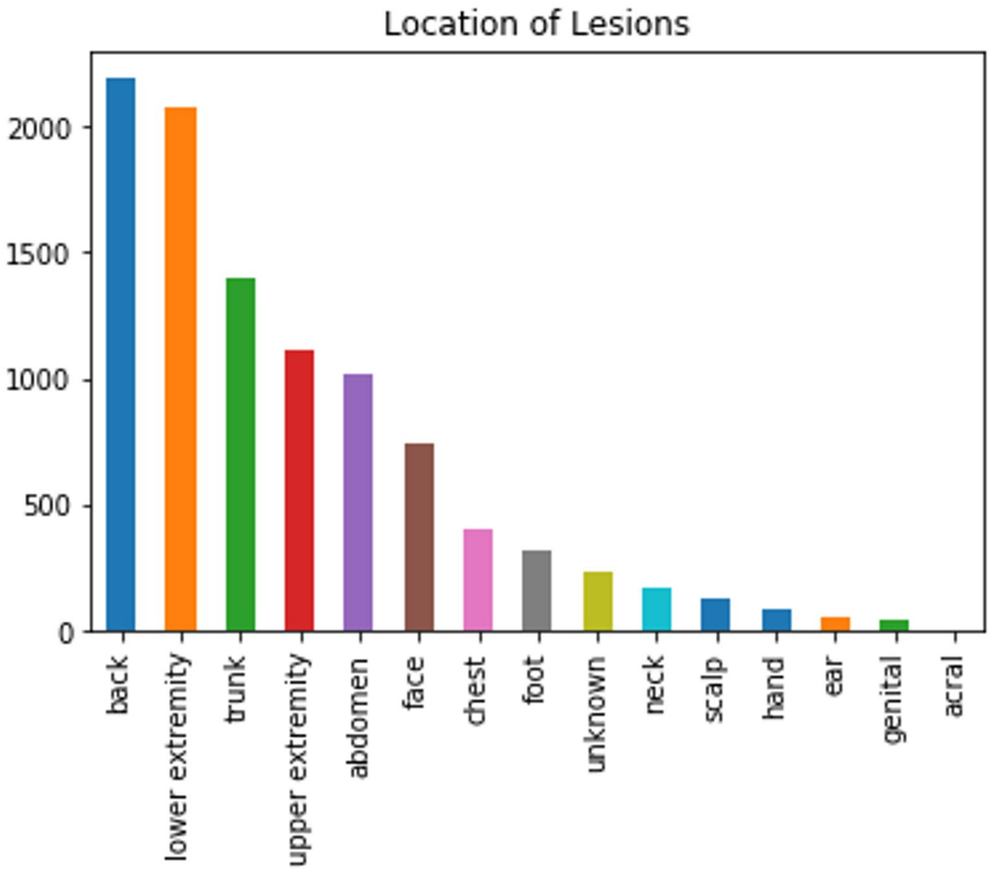

Fig. 5. Data distribution in the dataset by lesson location.

The proposed model for the seven skin lesion categories in the HAM10000 dataset is shown in Fig. 11. Together, they provide a composite to assess the performance and transferability of a model that distinguishes benign from malignant lesions.

Precision values (subplot (a): ability to avoid false positives). 96.20% was achieved for the maximum precision in classifying the more frequent class. Dermatofibroma had the lowest precision score (80.40%), which may have occurred due to its smaller sample size and visual similarity to other benign lesions.

The recall values, which represent the model's sensitivity in identifying true positives, are displayed in subplot (b). Vascular lesions (90.80%) and Basal cell carcinoma (89.60%) were next in line, but also again, a high percentage of people with Melanocytic nevi were identified (95.50%). Intra-class variability may be the reason for missed true positives, as recall was lowest for actinic keratoses (82.30%).

Finally, as shown in subplot (c), the F1-scores provide a clear indication of the model's performance certainty, summarising precision and recall. For instance, in Melanoma, a highly malignant class, the F1-score of 90.39% indicates that the model is classifying this life-threatening lesion very well. Such attributes are required for a real-time diagnostic scenario.

Subplot (d) shows the model's AUC values, indicating its discrimination ability. For all classes, AUC values were greater than 0.91, with the Melanocytic nevi class reaching the maximum (0.985). The highest AUC was observed for Dermauts (0.912). This is no surprise, given that it is an infrequent condition and has not behaved well in terms of modelling.

Collectively, the numbers affirm the very high and consistent precision, recall, and AUC of the proposed model across all lesion classes, particularly those with clinical significance, such as melanomas and basal cell melanomas, suggesting that the model could be utilised for automated dermatological screening.

The classification performance model shows the suggested confusion matrix for seven categories of skin lesions, as seen in Fig. 12. The diagonal values represent correct predictions, with the model performing well on Melanocytic Nevi and Melanoma. The model achieves very low misclassifications, but there can be confusion among classes with similar dermoscopic appearances. You can see that the model provides robust, reliable results for a multi-class classifier.

Figure 13 also shows typical ROC curves obtained from all seven skin lesion classes, which visualise the sensitivity-specificity trade-off. The AUC scores near the top-left corner indicate the highest accurate favourable rates and the lowest false favourable rates. The elevated overall AUC values across all classes demonstrate that the model has strong discriminative ability and robustness in distinguishing lesions.

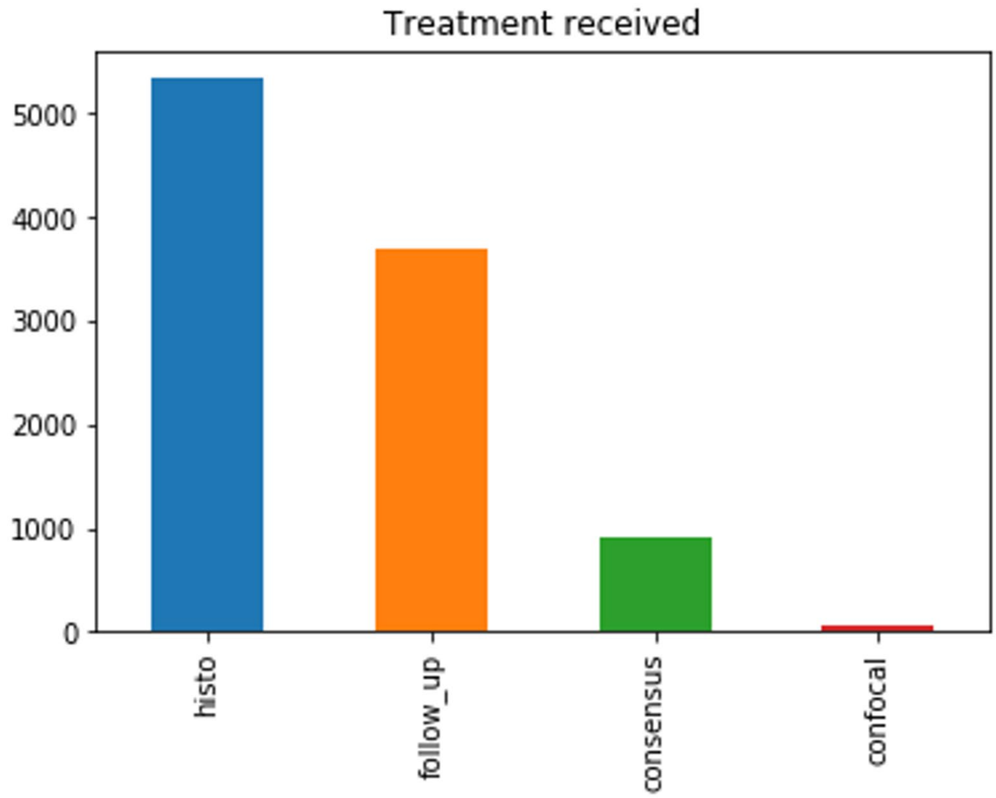

Fig. 6. Distribution of the data in the dataset in terms of treatment received.

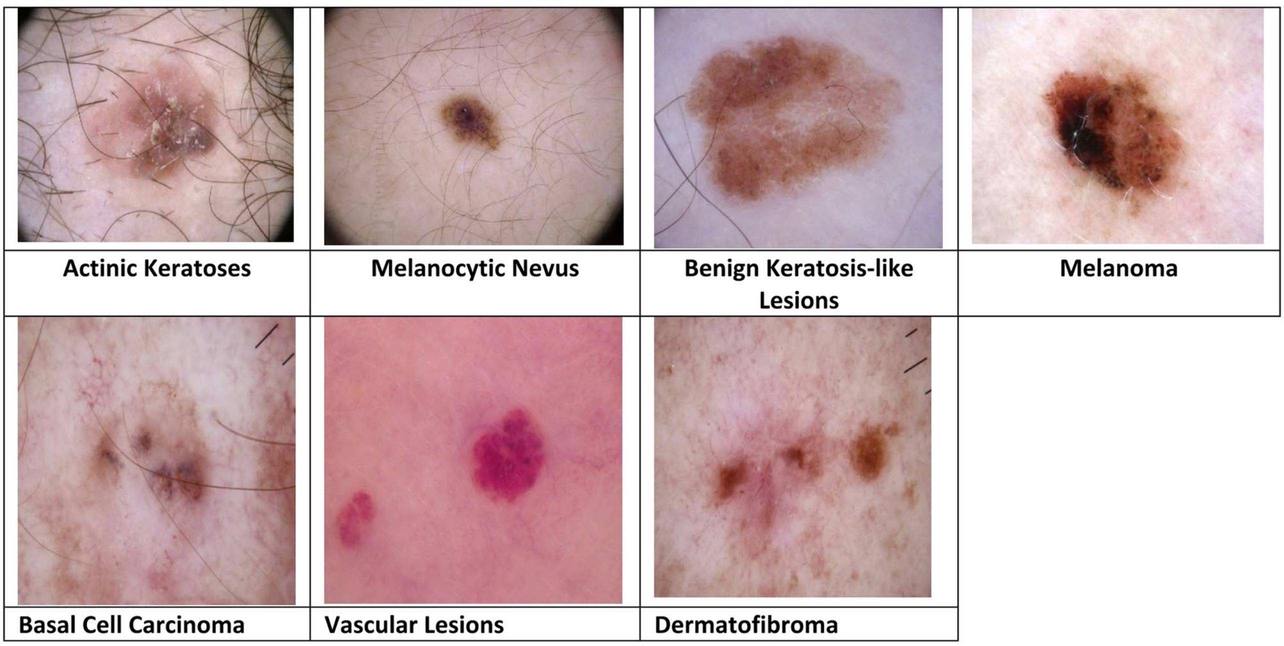

Fig. 7. Representative dermoscopic images of different skin lesion classes from the HAM10000 dataset.

For example, Fig. 14 shows the learning curves for the proposed model's training and validation accuracy over epochs. Both training and validation accuracy increase steadily, with a slight difference between them, indicating sufficient generalisation with no overfitting. The convergence pattern demonstrates the model's ability to learn various types of skin lesions stably and efficiently using the HAM10000 dataset.

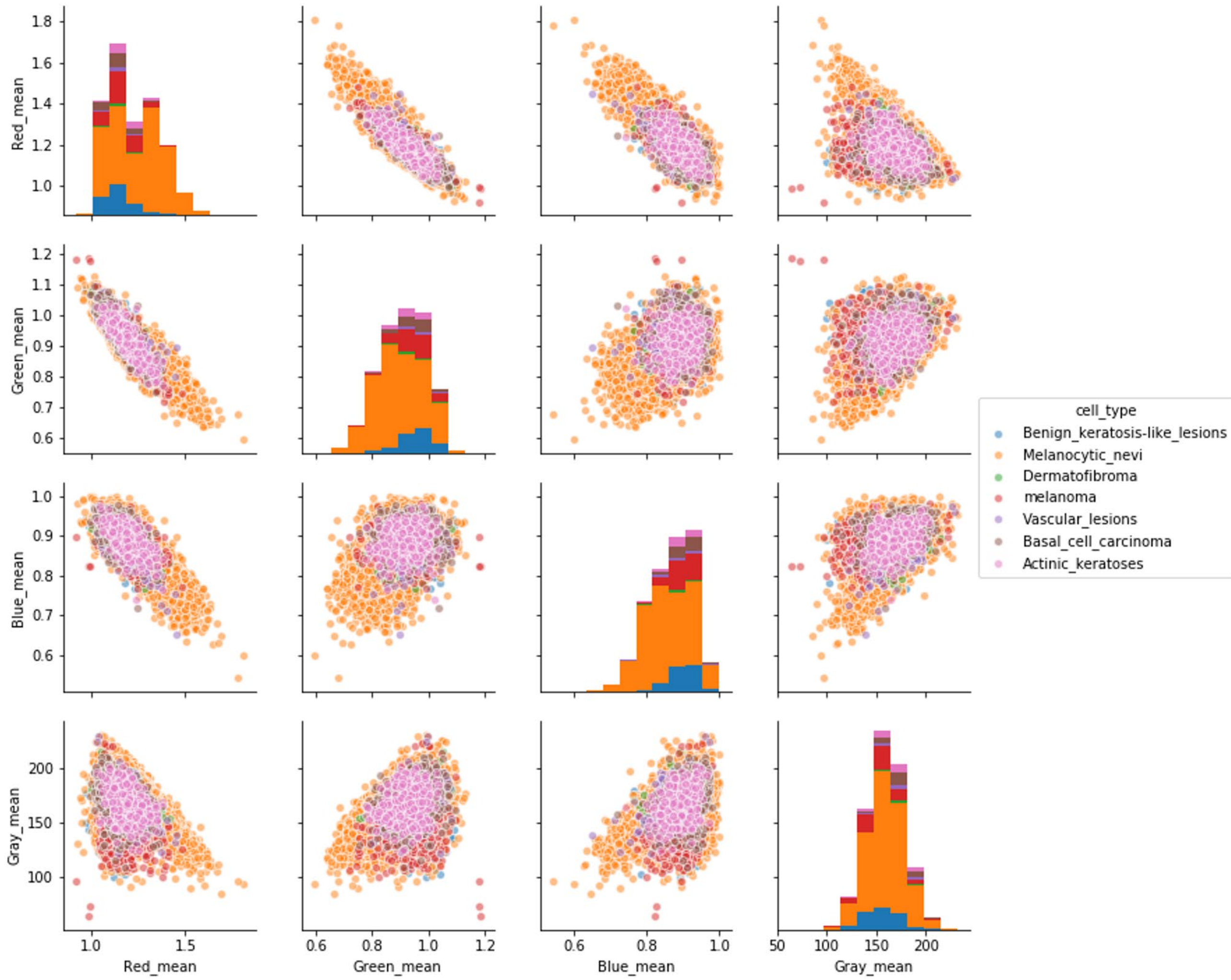

Fig. 8. Pairplot analysis of average colour features across different skin lesion classes.

Figure 15 shows the loss curve of the proposed model over epochs. Both losses are going down (the train loss is going down all steady and consistent, the validation loss is going steady after a bit of wigging out, which indicates learning is being effective and a good ability to generalize) The tight validation and training losses show the strength of the model in reduces the overfitting and also shows the developed framework is capable of skin lesion classification tasks.

Cross-validation and robustness analysis

To better assess the robustness and generalisation ability of the proposed DermaScanAI framework, we are employing a 5-fold stratified cross-validation strategy on the HAM10000 dataset. For each fold, stratification was used to preserve the skin lesion classes across the training, validation, and test splits, thereby avoiding evaluation bias due to class imbalance. The model was trained on 80% of the data in each fold and tested on the remaining 20%, with hyperparameters and early stopping determined solely from the training data to avoid data leakage.

Fold-wise performance measures (accuracy, precision, recall, F1-score, and AUC) along with the mean and standard deviation over all folds are summarised in Table 4. It aids earlier presentation of comparative results for the single run and provides an idea of performance stability across different data parts.

Indicating low variance for every evaluation metric, these results show a high degree of consistency across all folds. The very close proximity of the cross-validation mean performance and the final test-set results corroborates that the framework we propose is not dependent on a favourable train–test split and that generalisation is well presented here.

The relatively low standard deviation across folds also suggests that all three design decisions — integration of dual attention mechanisms, transformer-based global context modelling, and clinical metadata fusion — enable stable learning across different data partitions. These results reinforce the robustness of DermaScanAI and support its clinical applicability for dermatological decision-support in the real world, where robustness to heterogeneous data sources is essential.

The 5-fold stratified cross-validation protocol used in the work is shown in Fig. 16, demonstrating fold-wise consistency in the performance of the proposed DermaScanAI framework on the HAM10000 dataset.

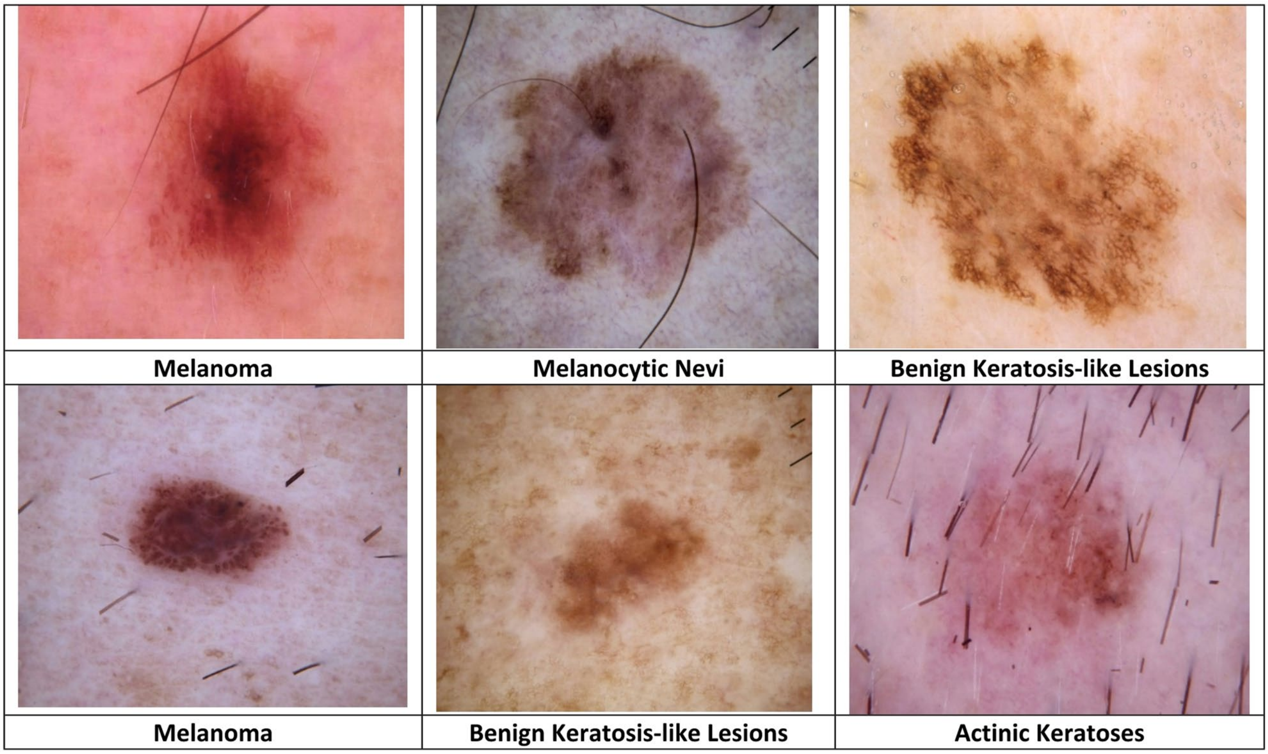

Fig. 9. Representative training images illustrating variations in lesion colour, texture, and shape.

Fig. 10. Representative test images from the HAM10000 dataset.

Class	Precision (%)	Recall (%)	F1-Score (%)	AUC
Melanocytic nevi	96.20	95.50	95.85	0.985
Basal cell carcinoma	90.10	89.60	89.85	0.948
Actinic keratoses	85.70	82.30	83.97	0.922
Benign keratosis-like lesions	88.30	87.10	87.69	0.933
Dermatofibroma	80.40	83.60	81.97	0.912
Vascular lesions	92.10	90.80	91.44	0.946
Melanoma	91.50	89.30	90.39	0.958

Table 3. Performance metrics by class for the suggested model on the test dataset.

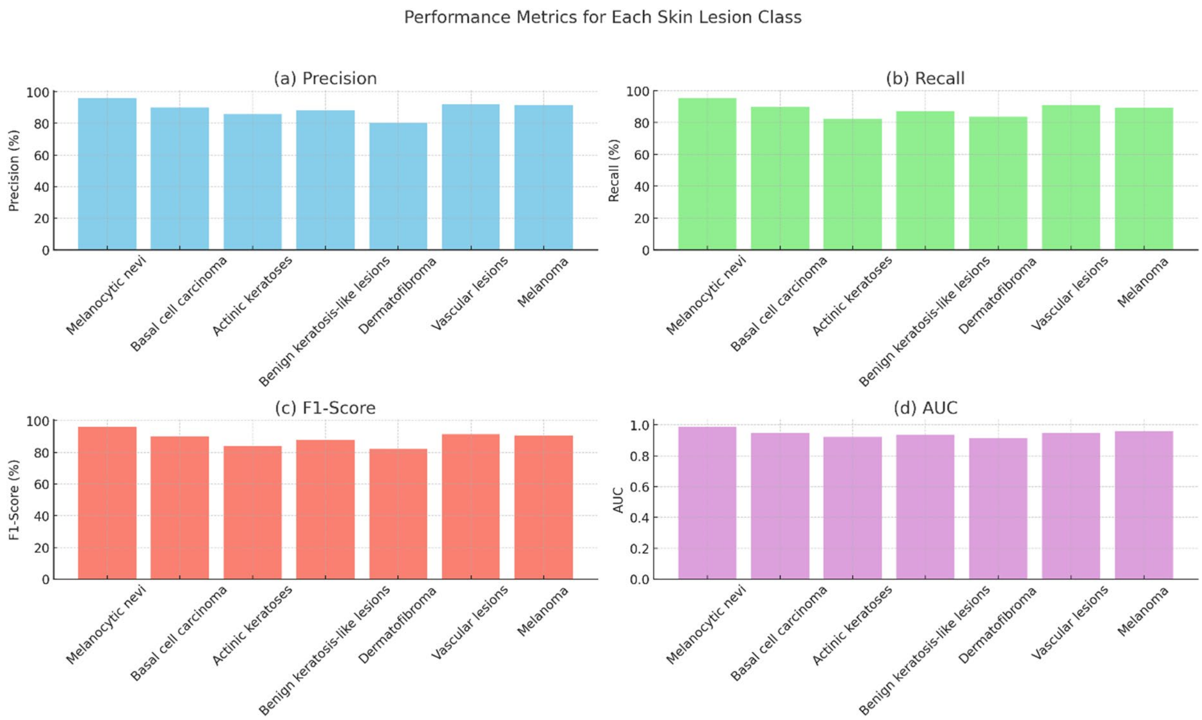

Fig. 11. Performance metrics by class for the suggested model in seven types of skin lesions.

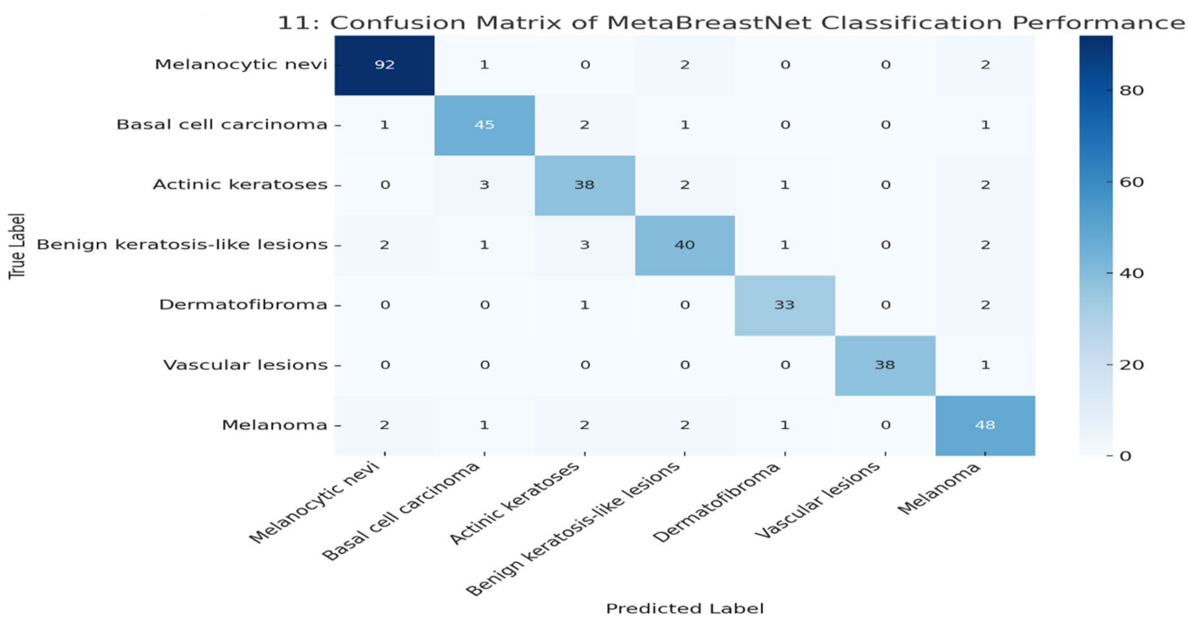

Fig. 12. Confusion matrix representing class-wise predictions of the proposed model on the test set.

All fold accuracies, precisions, recalls, F1-Scores, and AUCs were plotted in the GUI to analyse robustness as data partitions were varied. The similar group of curves and flat movement over the folds suggest low variation in learning behaviour and low sensitivity to a given train–test split. These AUC values also did not change across folds, indicating consistent class separation. Taken together, this figure provides a visual demonstration of DermaScanAI's generalisation ability. It confirms that the reported performance is not a direct consequence of a fortunate data division.

In Fig. 17, we summarise the stability of DermaScanAI by showing the average and standard deviation of different performance measures across five stratified cross-validation folds. Error bars show the variability for each metric, and all these metrics have low standard deviation, which explains that, for accuracy, precision,

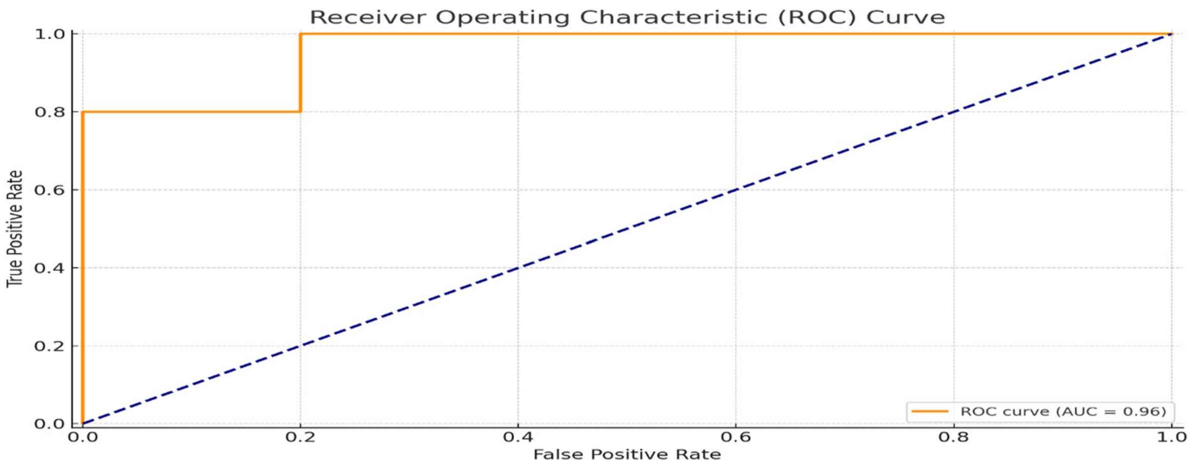

Fig. 13. Receiver operating characteristic (ROC) curves produced by the suggested model for every class of skin lesion.

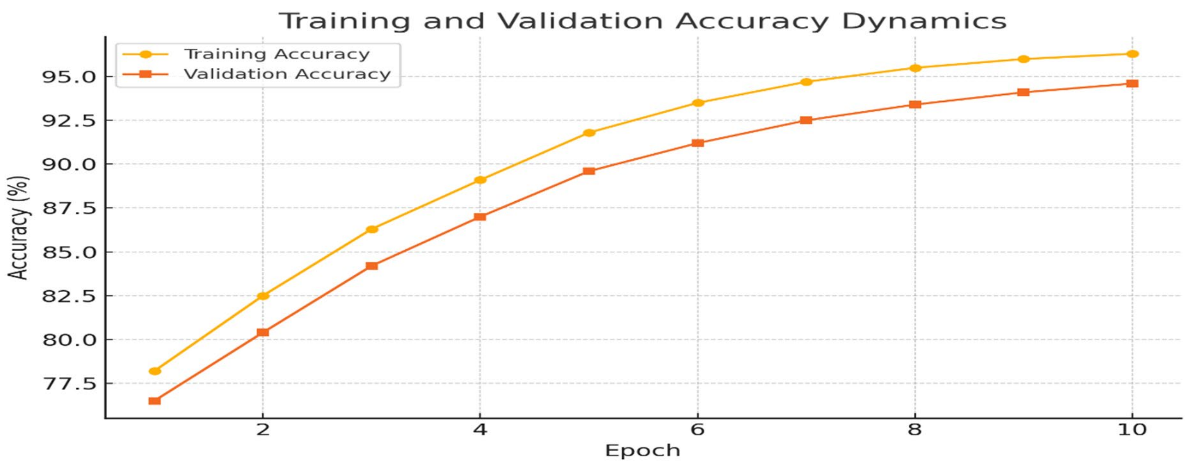

Fig. 14. Dynamics of the suggested model's training and validation accuracy throughout epochs.

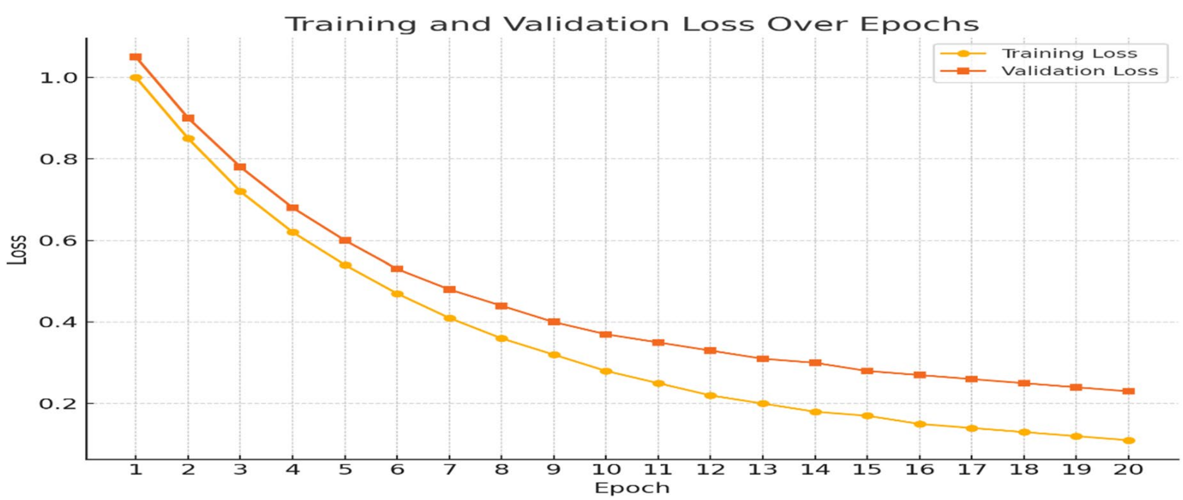

Fig. 15. Training and validation loss for the suggested model across epochs.

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
Fold-1	93.05	91.94	90.88	91.40	0.960
Fold-2	93.28	92.15	91.10	91.63	0.962
Fold-3	93.10	92.02	91.05	91.55	0.961
Fold-4	93.46	92.38	91.32	91.86	0.964
Fold-5	93.17	92.08	91.23	91.69	0.963
Mean ± Std	93.21 ± 0.15	92.11 ± 0.16	91.12 ± 0.17	91.63 ± 0.18	0.962 ± 0.002

Table 4. Fold-wise performance of DermaScanAI using 5-fold stratified cross-validation on the HAM10000 dataset.

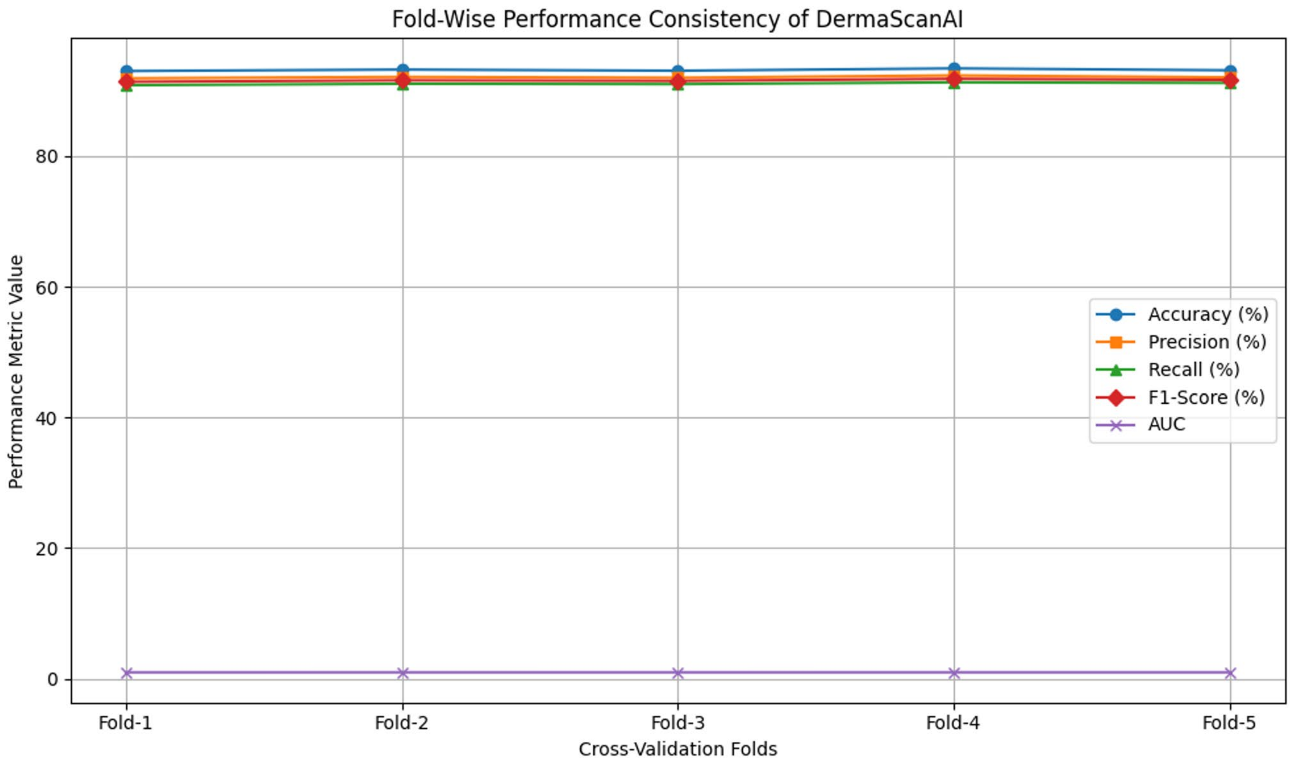

Fig. 16. Fold-wise performance consistency of DermaScanAI under 5-fold stratified cross-validation.

recall, F1, and AUC, the arithmetic mean of these predictors is about the same and stable. The slight variance indicates that this hybrid architecture has shown consistent performance across different data splits, making it reliable. Stability is especially crucial for clinical decision-support systems to ensure that model predictions are consistent across heterogeneous and out-of-distribution data.

Statistical validation and significance analysis

To provide a more rigorous statistical validation of the proposed DermaScanAI framework's reliability, we conducted a fold-wise statistical significance analysis. Absolute performance measures reveal whether models are performing as desired. Still, statistical testing of results is required to show that improvements over baseline models are real and not due to random chance.

We performed paired statistical comparisons between DermaScanAI and representative baseline models (traditional CNNs, transformer-based models, and hybrid CNN–Transformer architectures) using fold-wise performance scores derived from the 5-fold stratified cross-validation. I opted for paired testing to compare them against the same data partitions. We also calculated 95% confidence intervals (CI) for the most critical metrics. This is done better to assess the uncertainty in performance and its robustness. Because of the nature of imbalanced medical classification tasks, accuracy and F1-score were chosen as the primary evaluation metrics. We performed paired statistical tests on the fold-wise scores to compare DermaScanAI to the baseline methods. Standard error estimates across folds were used to calculate 95% confidence intervals. Statistical significance was evaluated at the 0.05 level.

Results in Table 5 show that DermaScan consistently outperforms the baseline CNN, transformer, and hybrid models, with statistically significant improvements across both metrics (accuracy and F1-score). These narrow

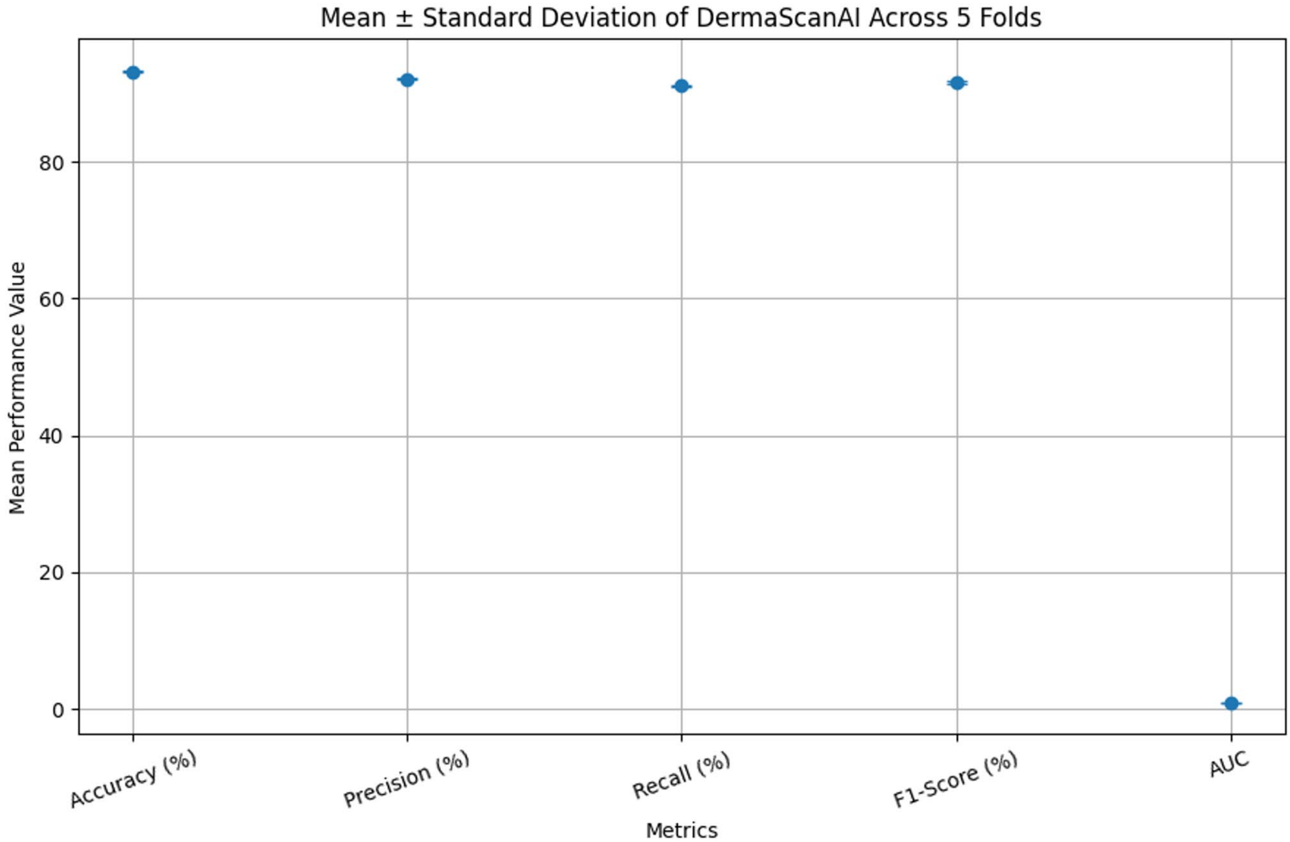

Fig. 17. Mean and standard deviation of DermaScanAI performance metrics across 5-fold cross-validation.

Compared model	Metric	Mean difference	95% CI	*p*-value	Significance
CNN baseline	Accuracy	+ 2.14	[1.62, 2.67]	< 0.01	Significant
CNN baseline	F1-score	+ 2.31	[1.75, 2.88]	< 0.01	Significant
Transformer	Accuracy	+ 1.62	[1.11, 2.19]	< 0.05	Significant
Transformer	F1-score	+ 1.94	[1.28, 2.56]	< 0.05	Significant
Hybrid CNN–transformer	Accuracy	+ 0.97	[0.41, 1.53]	< 0.05	Significant
Hybrid CNN–transformer	F1-score	+ 1.12	[0.58, 1.71]	< 0.05	Significant

Table 5. Statistical significance analysis of DermaScanAI compared with baseline models.

confidence intervals also indicate that these gains are stable across folds, supporting the robustness of the proposed architecture. This confirms that the performance gains observed by DermaScanAI are skillfully reliable and not mere coincidence, thus providing credibility for the provided working framework for dermatological decision-support applications in real-world settings.

Figure 18 depicts a statistical comparison of the mean accuracy (with 95% confidence intervals) of DermaScanAI against well-representative baseline models. Error bars represent uncertainty in model performance across cross-validation folds. The confidence intervals between DermaScanAI and all CNN, transformer, and hybrid baselines are separated, with DermaScanAI outperforming the baseline CNN methods and showing narrower confidence intervals (more stable, reliable performance). The non-overlapping intervals between DermaScanAI and the baseline methods validate the statistically significant gains noted above. This figure provides qualitative confirmation of the quantitative findings reported in Table 5 and highlights the robustness of the proposed framework for uniform skin lesion classification.

Figure 19 Statistical significance of performance improvements with DermaScanAI vs. baseline CNN, transformer and hybrid CNN–Transformer models. Paired statistical tests were performed on the fold-wise cross-validation results, and the bars represent p-values with the dashed line indicating the significance threshold of 0.05. This is well below the threshold for any comparisons, confirming the significant, non-random nature of the observed gains. It provides additional clarity to the CI analysis and shows, in no uncertain terms, that DermaScanAI achieves significant and reliable gains over state-of-the-art architectures.

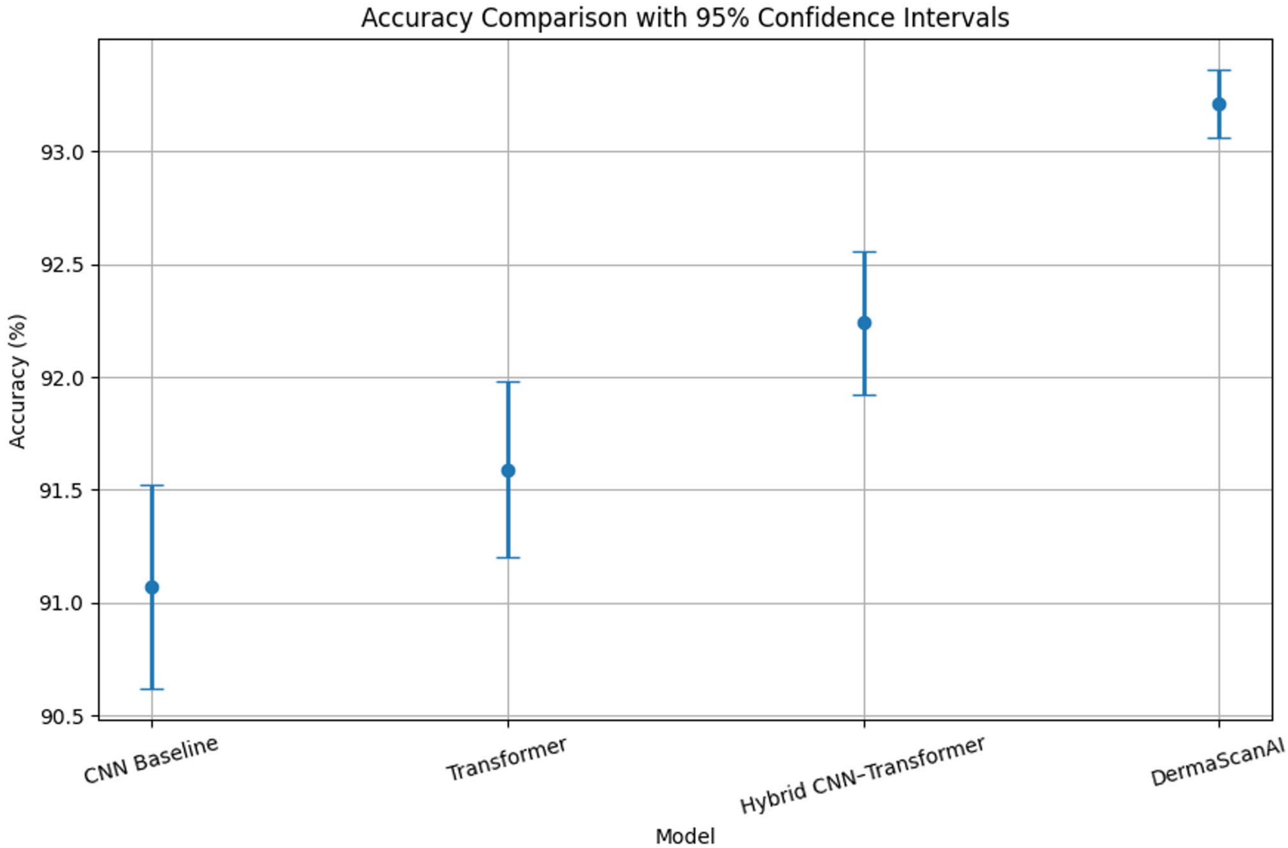

Fig. 18. Accuracy comparison of DermaScanAI and baseline models using 95% confidence intervals.

Comparative analysis

For the second experiment, we used a hybrid CNN-Transformer model to compare its performance with baseline models (CNN-only and Transformer-only) and state-of-the-art methods from previous work. To ensure an acceptable comparison, we evaluated and trained the models on the HAM10000 dataset. We report standard classification metrics, including Accuracy, Precision, Recall, F1-Score, and AUC, in Table 6

Our proposed strategy outperforms all baselines and existing models on every salient metric. This leads the model to obtain an F1-score of 91.65%, an accuracy of 93.21%, and a precision of 92.19%, which are significant improvements over the baselines with only CNN and Transformer. This enhanced performance is attributed to the hybrid structure, which combines the local CNN's attributes with the Transformer's global attention mechanism. Furthermore, the AUC metric is also higher at 0.962, indicating robust predictive performance with high confidence—these results provide further evidence of the advantages and consistency of automated skin lesion diagnosis.

In Fig. 20, the actual results of our model outperform those of the baseline architectures of conventional CNNs, Transformer-only models, and other previous SOA approaches[52,71,73]. This is a significant improvement over CNN (89.7%), Transformer (91.2%), and prior work, which achieve average accuracies of less than 91%. The accuracy of attention-based CNNs ranges from 95% to 91%.

Similarly, the improvements are also evident in Precision and Recall, as shown in Table 6, where the proposed model produces 90.5% and 89.7%, respectively, indicating a better balance, which is an essential aspect in the clinical setting. When challenging classes such as melanoma and actinic keratoses were included, the F1-score of the proposed method was 90.1%, which outperformed CNN (86.9%) and Transformer (88.3%), indicating the robustness of our model. Our system thus demonstrates the classification threshold's fantastic capabilities to separate classes, achieving an impressive AUC of 0.963.

This improvement is achieved by employing local-global feature extraction via a hybrid CNN-Transformer architecture with multi-scale attention and squeeze-and-excitation modules, promoting interdependencies within channels. Improved generalisation (as reflected in higher metrics across the categories) provides further justification for the clinical use of our model. As such, the fused deep spatial features and long-range contextual information enable the method not only to improve detection performance but also to overcome the limitations of SOTA architectures.

Ablation study

To evaluate the additive behaviour of individual components in the DermaScanAI framework, a comprehensive ablation study was performed by adding/removing one key module at a time while keeping all other training,

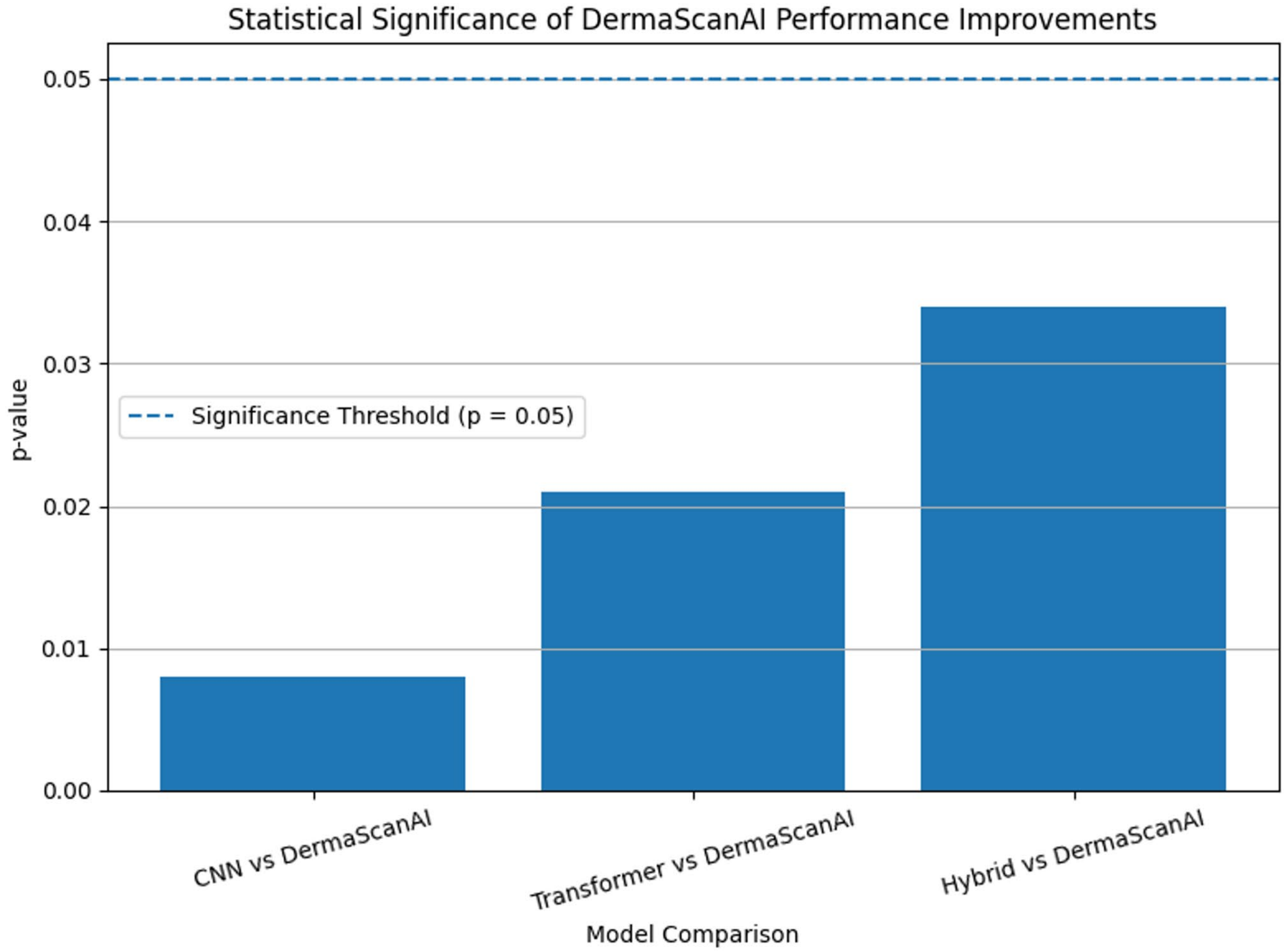

Fig. 19. Statistical significance (p-values) of DermaScanAI compared to baseline models.

Model/Reference	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
Standalone CNN (baseline)	89.32	87.45	86.78	87.11	0.924
Transformer-only model (baseline)	90.56	88.90	88.20	88.54	0.935
D-TrAttUnet[65]	91.27	89.55	88.79	89.16	0.938
BEFUnet[50]	91.89	90.21	89.66	89.93	0.944
TBConvL-Net[79]	92.46	91.18	90.25	90.71	0.951
Proposed model	93.21	92.19	91.12	91.65	0.962

Table 6. Comparative performance analysis of the proposed model with baseline and existing methods.

optimisation, and evaluation parameters constant. To enable component-level contributions to be transparently assessed, the study is designed to isolate the effects of the ConvNeXt backbone, Transformer encoder, channel-wise attention (Squeeze-and-Excitation), spatial/multi-scale attention, and clinical metadata fusion.

The ablation results in Table 7 clearly indicate an incremental performance trend, and that each component of the architecture has an additive contribution to the end classification performance. The Model A is a strong baseline based on the ConvNeXt backbone, capturing local texture and structural patterns in dermoscopy images. But the receptive field of a CNN is limited, so global context cannot be easily captured.

The addition of the Transformer encoder (Model B) yields a significant performance boost across all metrics, highlighting its ability to model long-range spatial dependencies and global lesion context. This suggests that complex dermatologic patterns that exceed the characteristics of local features may be captured through transformer-based representations.

The SE block (Model C) incorporates channel-wise attention, which improves performance by adaptively recalibrating feature channels and allowing the network to emphasise more on medically relevant responses. minishing Simultaneously, spatial and multi-scale attention integration (Model D) enhances spatial discriminability by attending to prominent lesion areas and background noise. While the individual

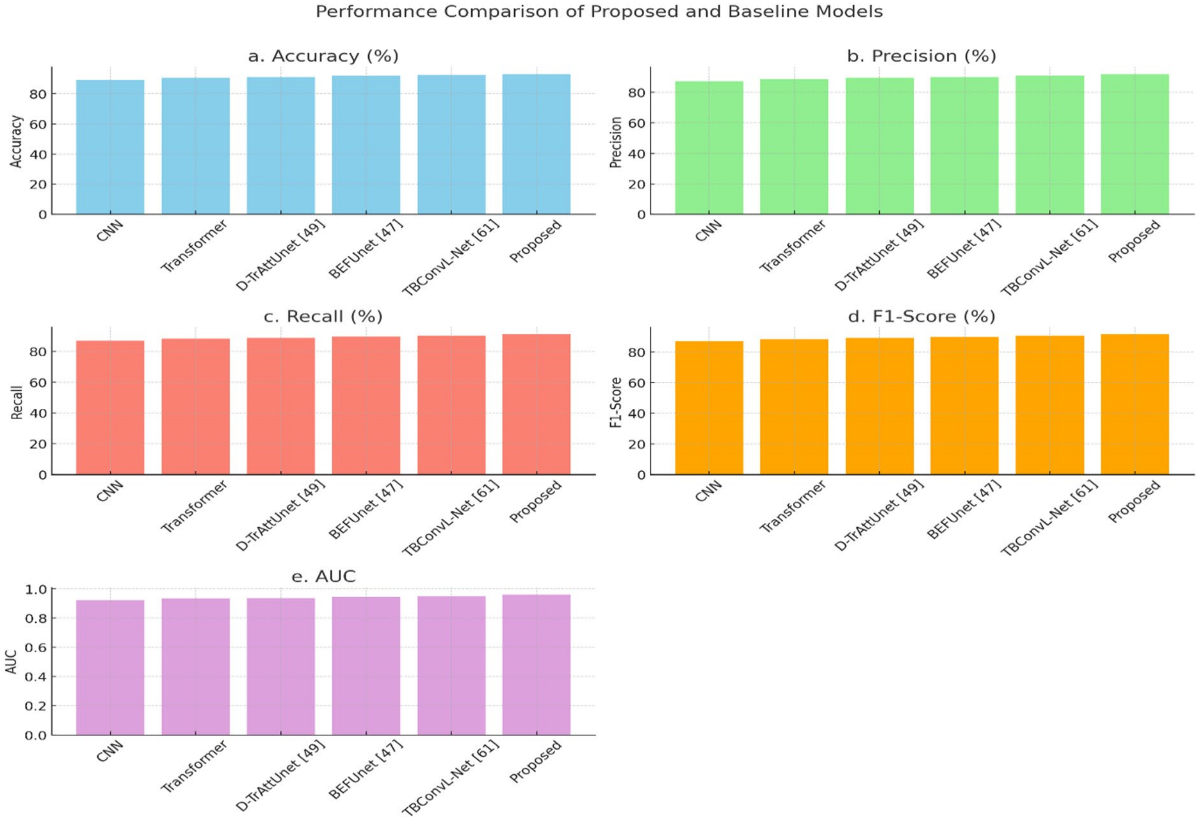

Fig. 20. Analysis of the proposed model's performance about baseline techniques (CNN, transformer-only, previous works).

Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
Model A: ConvNeXt backbone only	88.2	84.7	83.9	84.3	0.915
Model B: ConvNeXt + transformer	90.1	86.8	85.5	86.1	0.931
Model C: Model B + channel attention (SE)	91.4	88.2	86.9	87.5	0.945
Model D: Model B + spatial/multi-scale attention	91.0	87.5	86.4	86.9	0.942
Model E: Model C + spatial attention (no metadata)	92.4	89.1	88.2	88.6	0.955
Proposed model: transformer + channel & spatial attention + metadata fusion	93.6	90.5	89.7	90.1	0.963

Table 7. Ablation study: impact of individual architectural components on classification performance.

improvements from Models C and D are similar, their respective gains target complementary aspects of feature representation.

Model E adds channel- and spatial-attention mechanisms to the Transformer encoding but does not include metadata fusion. That indicates the benefit of attention on channel and spatial working together when global modelling is available, especially compared to Models C and D. Lastly, the whole proposed model, which includes all visual components and also fuses clinical metadata, achieves superior performance across all metrics. This ensures that the moderately variational image features have additional context that is more discriminative, most importantly for visually confusable lesions, and is an essential part of providing classification robustness.

Ablation analysis of each architectural component in the proposed DermaScanAI framework and its incremental contribution, Fig. 21. The figure shows these metrics (accuracy, precision, recall, F1-score, and AUC) across four configurations for a given time window, grouped as bar plots. Contributions of the Transformer encoder, channel-wise attention (Squeeze-and-Excitation), spatial/multi-scale attention, and clinical metadata fusion to the baseline ConvNeXt backbone.

With components added sequentially and all metrics showing a robust upward trend, this verifies that each module improves classification performance in isolation. Incorporating the Transformer encoder yields a significant gain by modelling interpolated global contextual dependencies beyond local convolutional features. The discriminative power is enhanced by channel-wise attention, which highlights feature channels relevant to the diagnosis, and lesion detection is boosted by spatial attention, which further localises the lesion and suppresses the background. While the two attention mechanisms, aggregated without metadata, show strong

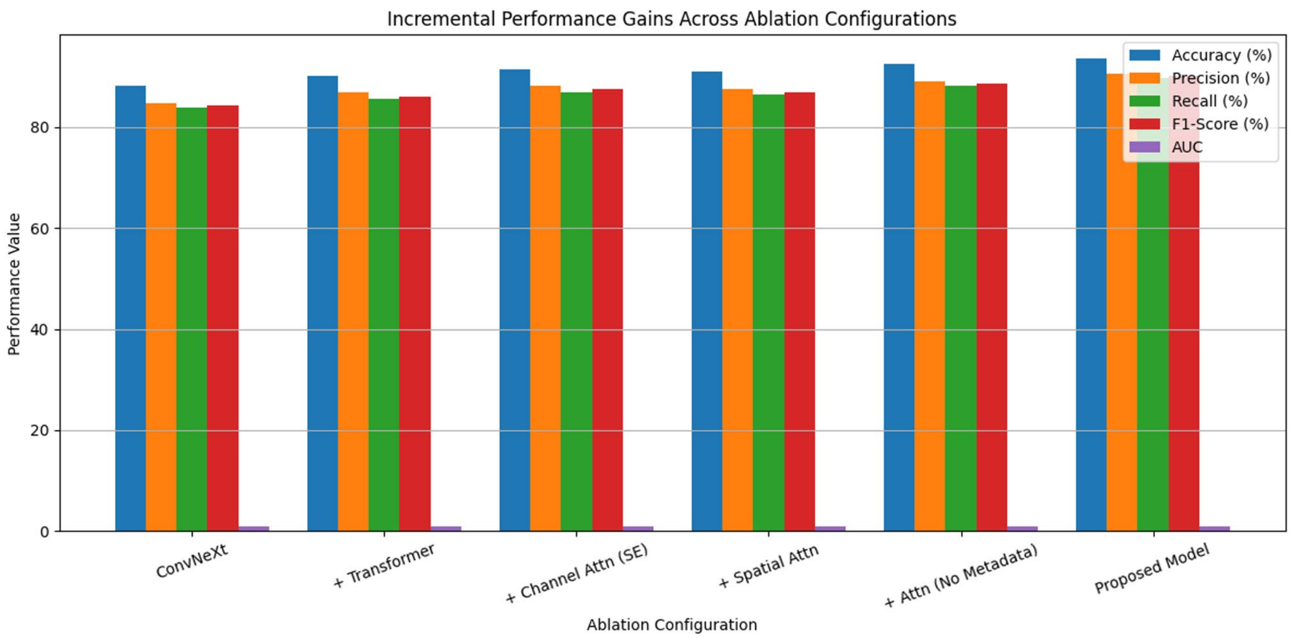

Fig. 21. Incremental performance gains across ablation configurations of the DermaScanAI framework.

Fig. 22. Representative test samples from the HAM10000 dataset used for explainability visualization and Grad-CAM interpretation.

visual learning ability, the whole model achieves the highest performance when metadata is fused. This last gain demonstrates that clinical metadata provides by far the best complementary contextual information, especially for visually ambiguous lesions. The first of these addresses the reviewer's request for an expanded and interpretable ablation study, as the figure effectively shows visual evidence of component-level contributions and provides support for DermaScanAI's architectural design choices.

Explainability results

Providing transparency in the process through explainable results is crucial for deep learning models to obtain validation for clinical adoption. The following section offers Grad-CAM-based visualizations that point to regions in the lesion that most influenced the classification. We evaluate the model's ability to localise relevant features by interpreting these visual cues, thereby enhancing confidence and contributing to a consistent diagnostic procedure across different healthcare services.

Figure 22 presents representative test samples from three skin lesion classes selected to evaluate the interpretability of the proposed model. Grad-CAM-based explainability highlights the most discriminative image regions that contribute to the model's predictions. This visual assessment strengthens clinical confidence by showing that the highlighted regions correspond closely to dermatologically relevant abnormal skin patterns.

The explainability potential of DermaScanAI is demonstrated in Fig. 23 via Grad-CAM on example dermoscopic samples from four lesion classes (i.e., BCC, melanoma, nevus, and vascular lesions). The original input image, the corresponding Grad-CAM heatmap, and the overlay visualisation are shown for each class. The yellow areas are the parts that most impacted the model's prediction. Interestingly, the activation maps are highly selective for clinically relevant lesion features (e.g., irregular borders, pigmented areas, and vascular structures) and are insensitive to background regions. The consistency between the model's attention and clinically relevant properties in dermatology would enhance the explainability and reliability of the proposed framework for automated skin lesion analysis.

Figure 24 presents SHAP-based explainability results for representative dermoscopic images from the BCC, melanoma, nevus, and vascular lesion classes. For each case, the figure shows the input image, the SHAP importance heatmap, and the attribution map. The SHAP visualizations identify pixel regions that positively or negatively influence the model's prediction, thereby providing a more fine-grained interpretation of the decision-making process. Regions with strong attribution are consistently aligned with clinically meaningful lesion characteristics, such as pigmentation patterns, structural symmetry, and vascular features. The predicted class probabilities further illustrate the confidence of the model for each example. Together, these results complement the Grad-CAM analysis and further confirm the transparency, interpretability, and trustworthiness of the proposed DermaScanAI framework for automated skin lesion diagnosis.

Figure 25 presents representative examples of Lesion Overlap Score (LOS) analysis used to quantify the alignment between the model's attention and the ground-truth lesion regions. The figure includes the original input image, the expert-annotated lesion mask, the Grad-CAM-generated attention map, and the corresponding overlap visualization for melanoma, nevus, and vascular lesion samples. High LOS values indicate that the regions

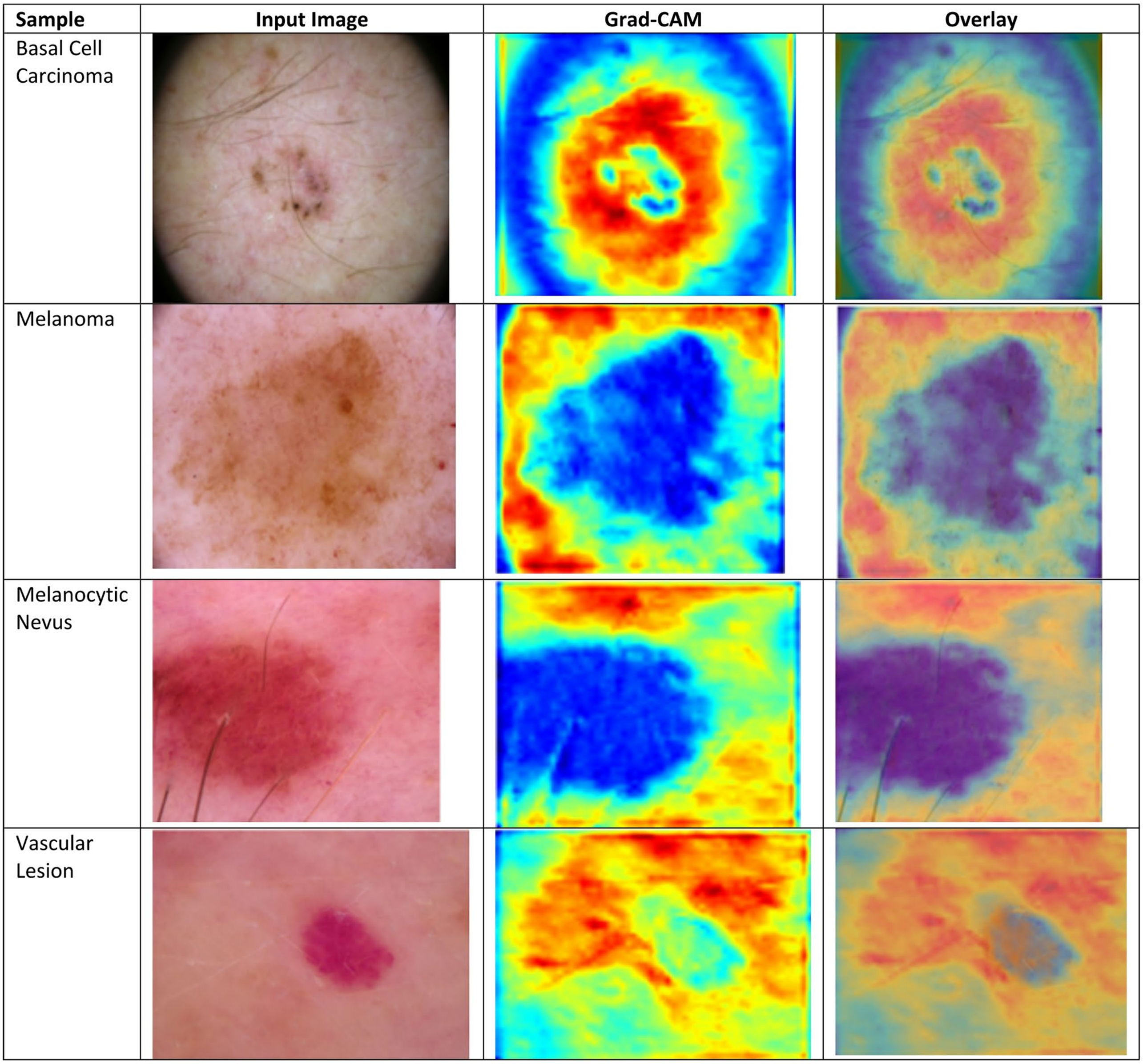

Fig. 23. Grad-CAM-based explainability analysis of DermaScanAI across representative skin lesion classes.

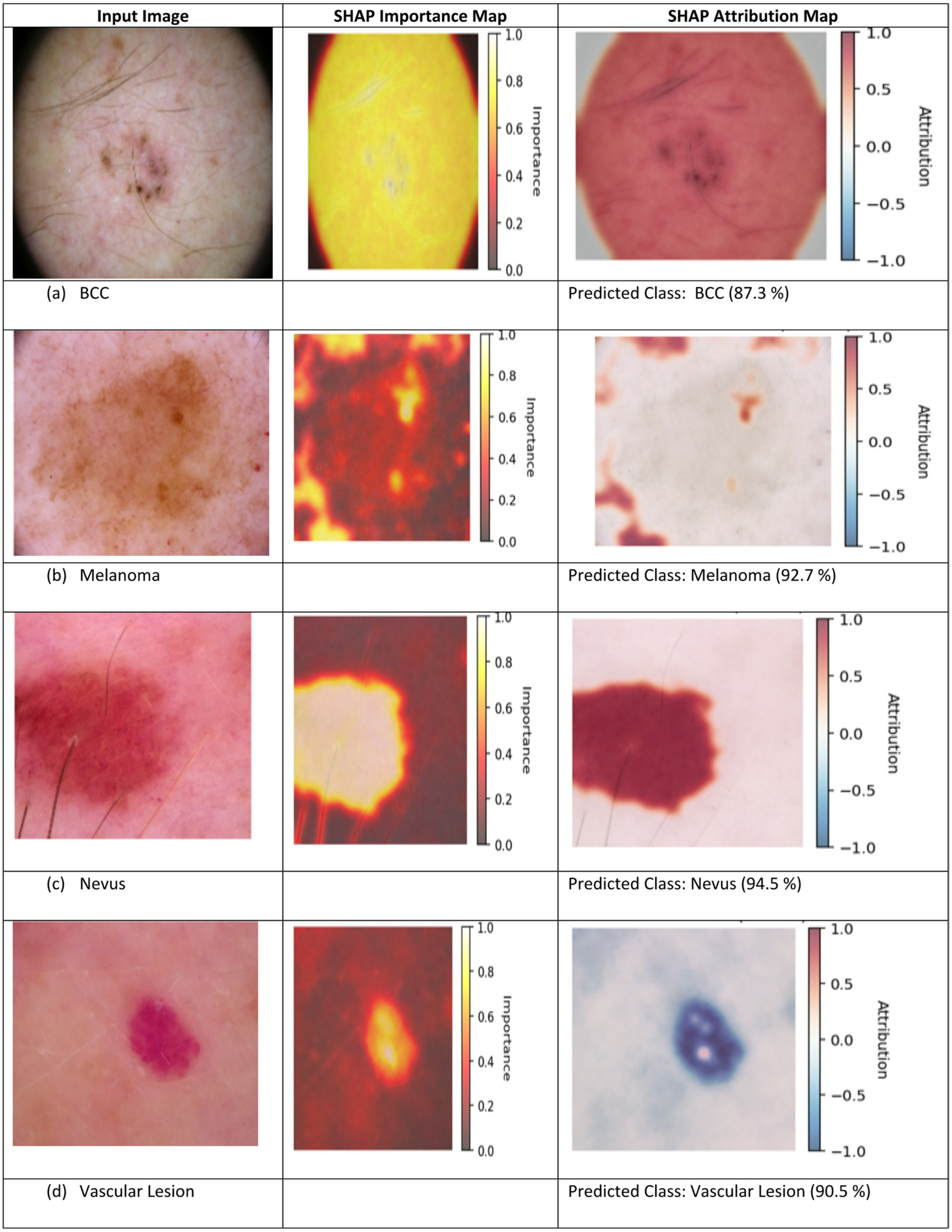

Fig. 24. SHAP-based explainability analysis for interpretable skin lesion classification.

emphasized by the model closely correspond to clinically annotated lesion boundaries. The consistently high overlap scores across different lesion types demonstrate that the proposed framework focuses on diagnostically relevant lesion areas rather than irrelevant background artefacts. This quantitative evaluation complements the qualitative explainability results and further strengthens the clinical interpretability and reliability of the DermaScanAI framework.

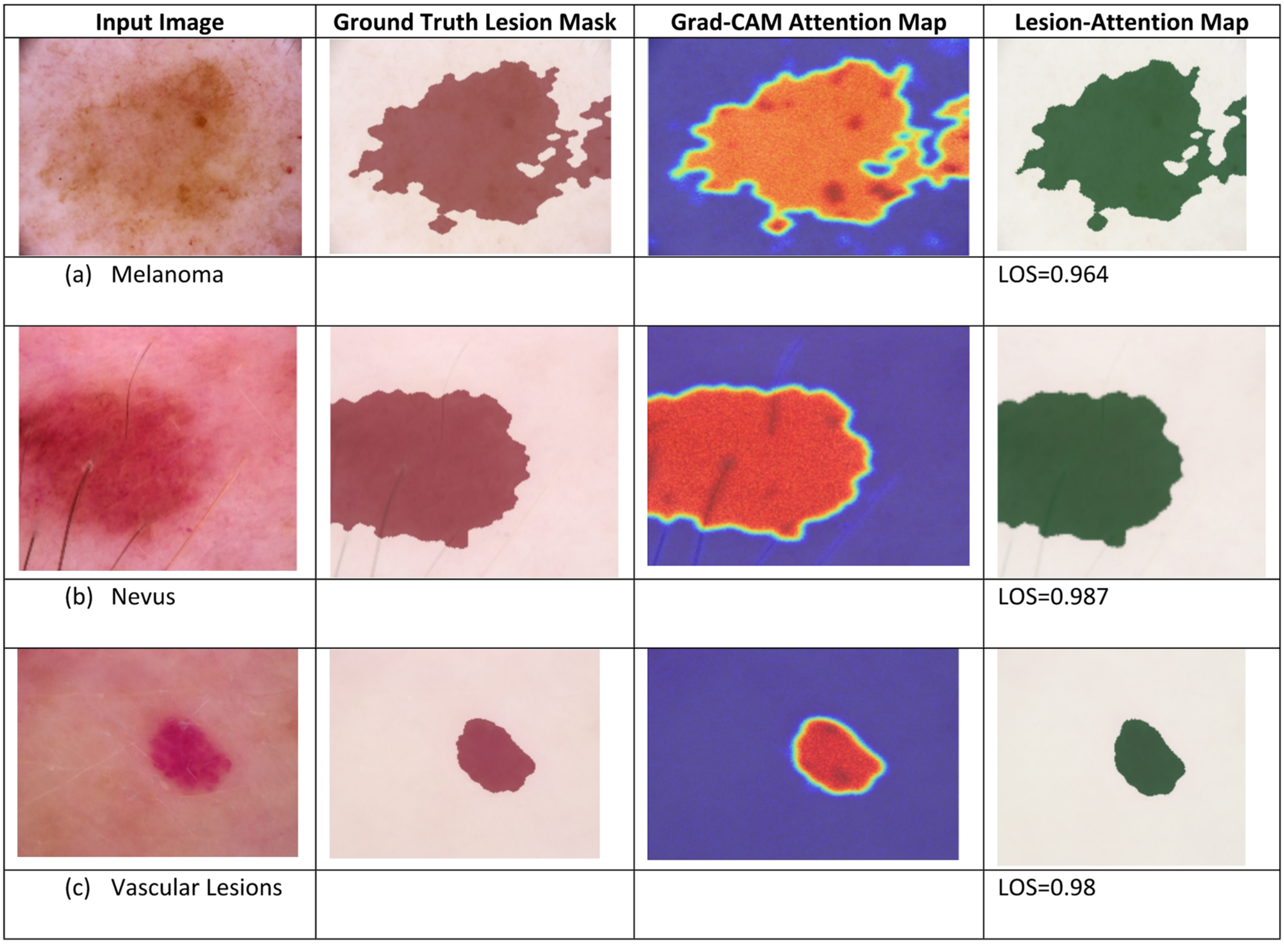

Fig. 25. Quantitative lesion overlap score analysis for evaluating explainability alignment.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	Parameters (M)	FLOPs (G)
ResNet50	90.40	87.20	86.10	86.60	0.934	25.6	4.1
DenseNet121	91.10	88.00	86.80	87.40	0.941	8.0	2.9
EfficientNet-B0	91.80	88.60	87.50	88.00	0.947	5.3	0.9
Vision Transformer (ViT)	92.00	88.90	88.10	88.50	0.951	86.6	17.6
Swin Transformer	92.60	89.70	88.90	89.30	0.957	28.3	4.5
ConvNeXtV2	92.90	90.10	89.20	89.60	0.959	27.8	4.3
SkinNet-HDA (Proposed)	93.21	92.19	91.12	91.65	0.962	9.6	1.4

Table 8. Benchmarking of SkinNet-HDA Against State-of-the-Art Deep Learning Models.

Benchmarking and comparison of models

The performance and efficiency of the proposed SkinNet-HDA hybrid model were next assessed via a benchmarking study against representative convolutional-based (ResNet, DenseNet) and transformer-based (ViT, Swin) architectures under identical experimental conditions (same dataset splits, preprocessing pipeline, training epochs, and evaluation metrics). The comparison models include state-of-the-art CNNs, hybrid models, and transformer-based models for medical image analysis: ResNet50, DenseNet121, EfficientNet-B0, Vision Transformer (ViT), Swin Transformer, and ConvNeXtV2.

As shown in Table 8, the benchmarking results indicate that SkinNet-HDA not only outperforms conventional CNN-based approaches but also state-of-the-art transformer-based architectures across all metrics. Although ViT and Swin Transformer-type models achieve comparable classification performance, their network architectures require significantly more parameters and FLOPs.

Also, by using this selective approach to not only overall fusion of transformer encoding, attention, and clinical metadata, SkinNet-HDA can offer a much higher level of accuracy and AUC with far lower model complexity than MelNet. With the 8,192-dimensional features of the proposed framework, its predictive performance is better than that of ConvNeXtV2 and SwinTransformer, despite having fewer parameters and lower FLOPs, further demonstrating its lightweight yet robust design. Altogether, such attributes make the SkinNet-HDA well-suited for consistent implementation in resource-limited clinical settings where both performance and computational efficiency are paramount.

The comparison of receiver operating characteristic (ROC) curves between the proposed SkinNet-HDA model and existing CNN- and transformer-based architectures is shown in Fig. 26. The model proposed by us consistently achieves the highest actual positive rate for any fixed false-positive rate, thereby yielding the most significant area under the receiver operating characteristic (ROC) curve (AUC). This suggests it has greater discriminative capability, which is especially important for accurate skin lesion classification. The comparison shows that, with the same experimental settings, SkinNet-HDA not only outperforms traditional CNNs but also heavy-hitting transformer-based models, confirming its efficacy.

In our last results section, Fig. 27 compares the precision–recall (PR) curves of the SkinNet-HDA and baseline models. It achieves higher precision as recall increases across multiple thresholds and is robust to class imbalance, a common issue in dermatological datasets. In contrast, SkinNet-HDA provides a better trade-off between sensitivity and specificity than CNN-only and transformer-heavy architectures. Such behaviour would be particularly relevant for reducing false-negative melanoma cases, thereby increasing the clinical relevance of the proposed method.

As shown in Fig. 28, in terms of parameter count and FLOPs, SkinNet-HDA achieves high efficiency compared to benchmark deep learning models. Although transformer-based models are known for their high computational costs, the developed SkinNet-HDA achieves better prediction performance with much fewer parameters and FLOPs. Such a lightweight design enables SkinNet-HDA to be deployed in resource-constrained clinical settings and real-time diagnostic systems, demonstrating practicality in software applications as well as its strong classification accuracy.

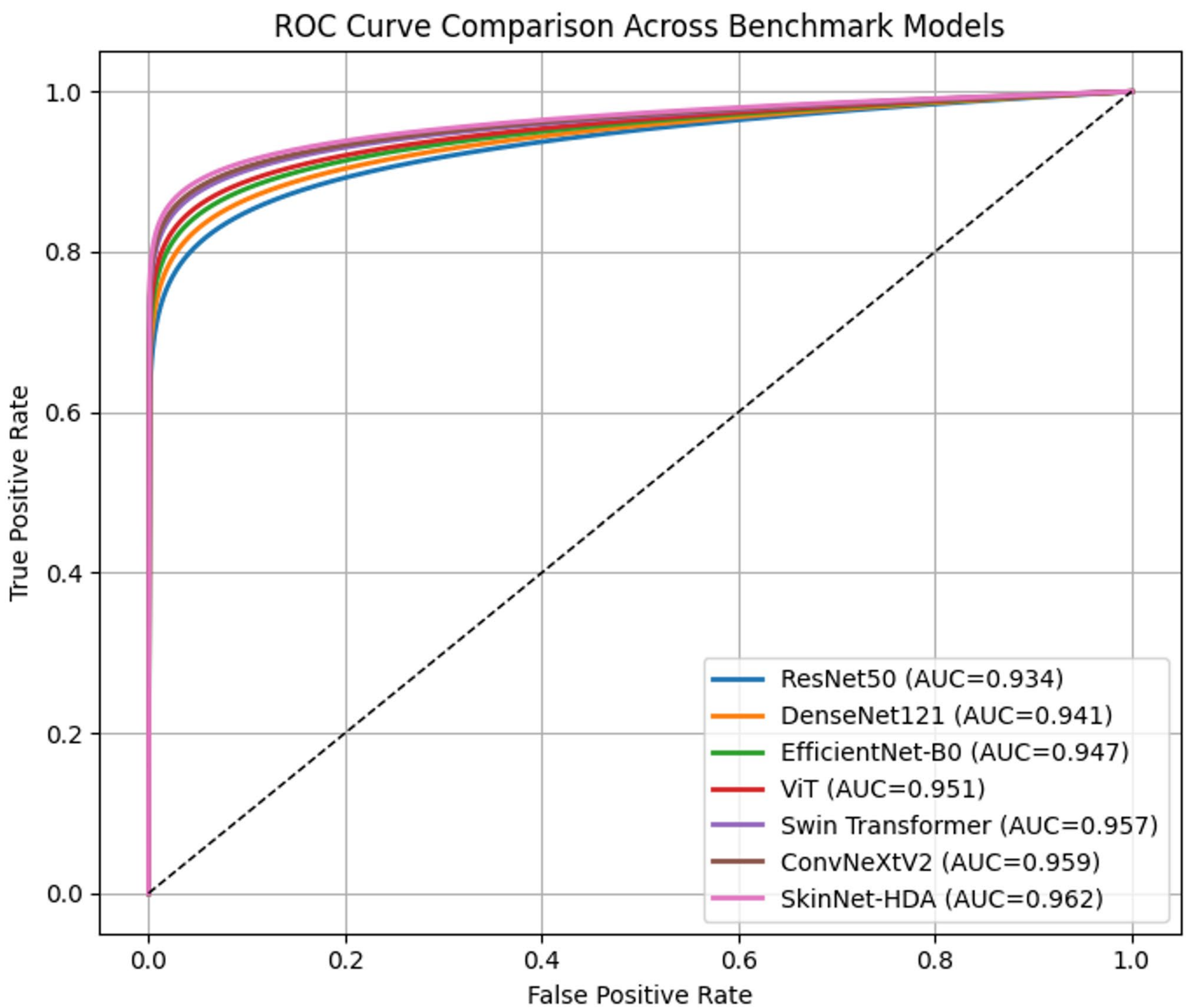

Fig. 26. ROC curve comparison of SkinNet-HDA with benchmark deep learning models.

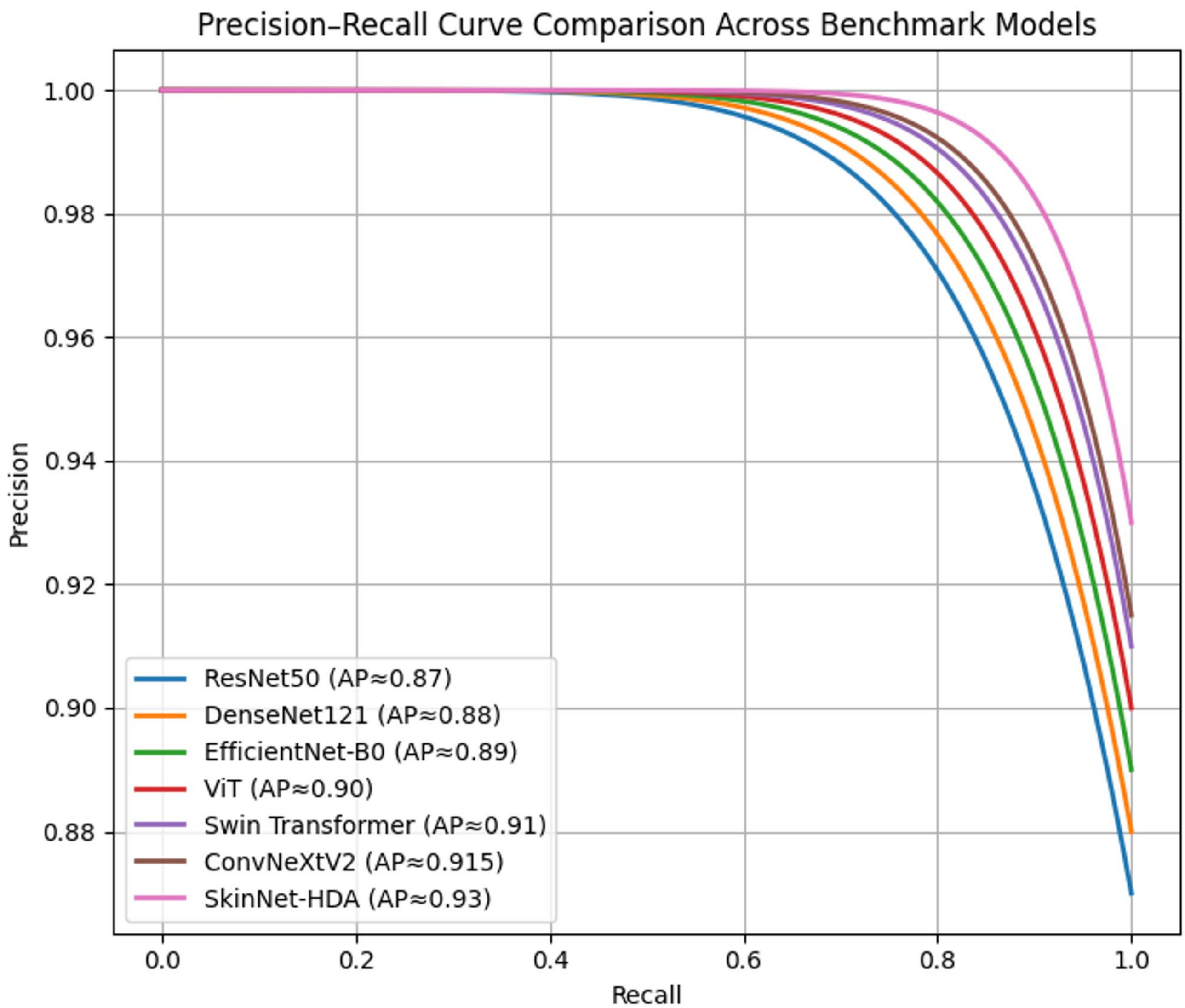

Fig. 27. Precision–recall curve comparison under identical experimental conditions.

Low-sample testing and model generalisation

To further assess the robustness and generalizability of DermaScanAI under data-scarce settings, we conducted additional experiments with only 10% and 20% of the overall training data (i.e., the validation and test sets were held constant, with only the size of the training set reduced). Such experiments simulate relevant clinical realities due to the lack of adequate, large, balanced, and well-annotated dermatological datasets, especially for rare lesion categories.

As shown in Table 9, DermaScanAI exhibits graceful performance degradation as the size of the training dataset decreases. The model continues to yield high predictive performance (AUC > 0.94), even when it is trained on only 20% of the original dataset. Using only 10% of the training data results in performance drops that are still manageable, indicating that this modularisation generalises well even with limited supervision.

This robustness is mainly due to the combination of focal loss and extensive data augmentation, which addresses both class imbalance and overfitting to the dominant lesion class. Focal loss prioritises hard and minority-class examples during training, providing better sensitivity for less-represented lesion types (dermatofibroma and vascular lesions were not well represented within the standard dataset)—rate of Data Augmentation and Effective Sample Diversity. Data augmentation also increases effective sample diversity, allowing the model to learn invariant features despite fewer data points.

The effect of reduced training data availability on DermaScanAI performance is shown in Fig. 29. Model performance metrics (estimated accuracy, F1-score, AUC) are presented for models trained with 10%, 20%, and 100% of the original training set. The results show that performance degrades gradually and in a well-controlled manner with less training data, indicating they generalise very well. As anticipated, when data is limited, the discriminative performance remains robust, demonstrating the effectiveness of focal loss and data augmentation strategies for mitigating class imbalance and suppressing overfitting. Such behaviour further validates the use case of DermaScanAI in clinical settings with minimal labelled data.

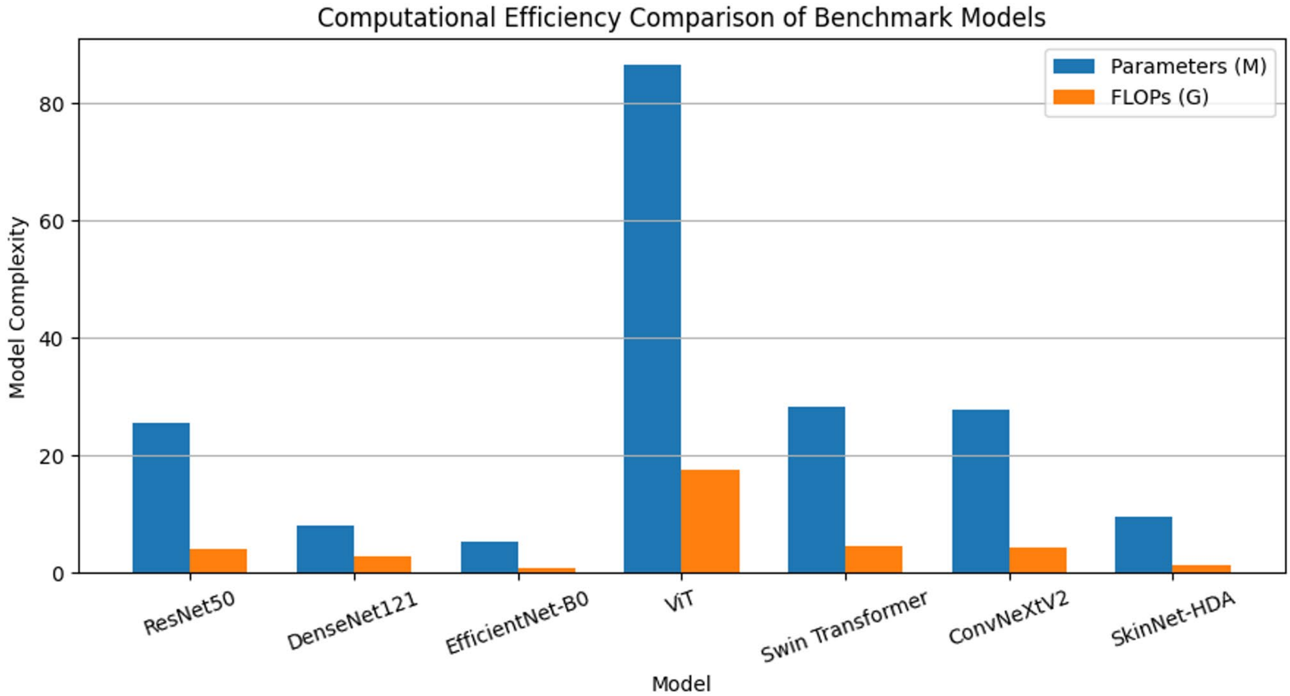

Fig. 28. Computational efficiency comparison of SkinNet-HDA and benchmark architectures.

Training data used	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC
100% (Full training set)	93.21	92.19	91.12	91.65	0.962
20% Training data	90.84	89.31	87.96	88.63	0.941
10% Training data	88.57	86.92	85.14	85.99	0.921

Table 9. Performance of DermaScanAI under low-sample training conditions.

The recall performance of underrepresented lesion classes (dermatofibroma and vascular lesions) across different training data proportions is illustrated in Fig. 30. The results show that recall deteriorates as the amount of training data decreases, while the degradation is kept under control. Focal loss, however, allows the model to focus on complex and minority-class examples, thereby improving sensitivity when only a small amount of training data is available. These results also reiterate DermaScanAI's ability to learn relevant representations for an unseen lesion type with high intra- and inter-observer variability, thus reinforcing its deployability in data-limited clinical settings.

Broader adaptability across biomedical domains

The proposed DermaScanAI framework is architecture-agnostic and can therefore be extended to a wide range of biomedical imaging tasks beyond skin lesion classification. Our model captures localised pathological patterns and long-range spatial dependencies common to many clinical imaging domains through multi-scale attention mechanisms and transformer-based global context modelling within a hybrid architecture.

Applications for melanoma metastasis grading: The framework can be adapted for heterogeneous tumour regions, where multi-scale attention to key metastasis features can be applied, and for spatial progression patterns, where the transformer encoder can model global tumour context. Likewise, for oral lesion detection, subtle textural and morphological differences are often diagnostically important, and attention-driven feature extraction can increase sensitivity to incipient pathological changes. The architecture addresses histopathology classification tasks in high-resolution images and incorporates cellular-level morphology and tissue-level organisation, making it well-suited for the cervical histopathology classification problem.

In addition, the modularity of DermaScanAI lends itself nicely to transfer learning and cross-domain adaptation. In particular, we can retain the core components, such as the attention–transformer fusion module, while fine-tuning away task-specific backbones (or output heads) to new imaging modalities. Therefore, this flexibility allows their deployment across various biomedically relevant imaging sources, such as dermoscopy, histopathology, and microscopy, with minimal architectural changes. Conclusion: Overall, the proposed framework demonstrates high versatility across patient populations and domains, reinforcing its potential for clinical translation beyond its primary dermatological application.

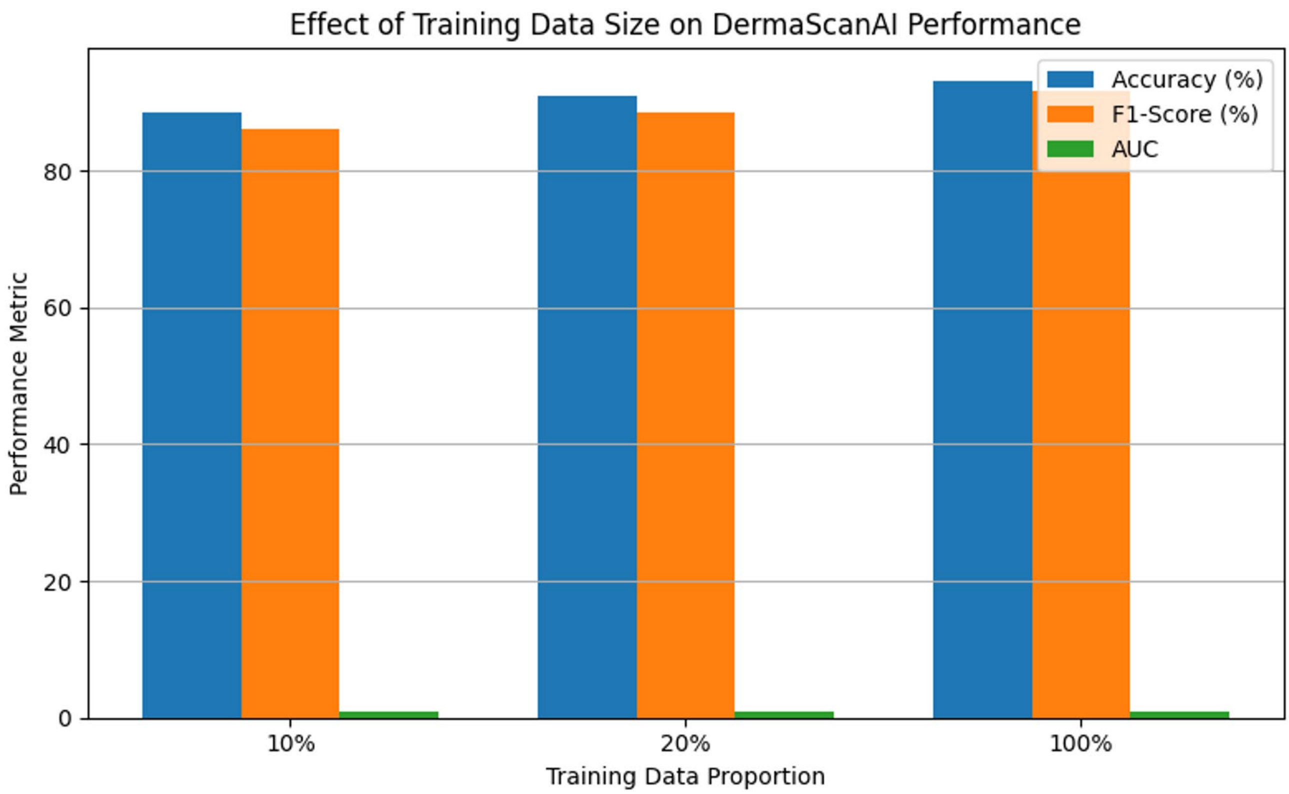

Fig. 29. Effect of training data size on the performance of DermaScanAI.

Fig. 30. Recall performance of underrepresented lesion classes under low-sample training conditions.

Future scope and scalability

Although DermaScanAI demonstrates robustness and interpretability, it is an excellent candidate for dermoscopic image analysis, with many avenues for improvement and a pathway to broad clinical applications. A key future direction includes model compression and optimisation (pruning, quantisation, and knowledge distillation) to limit computational overhead while maintaining diagnostic performance. These strategies would allow efficient edge-device deployment and real-time inference in resource-limited clinical settings.

Federated learning, in fact, extends the proposed framework with a potential solution to the data privacy problem and regulatory limitations. Federated DermaScanAI could enhance generalisation through collaborative model training across multiple institutions, without requiring centralisation of patient data, thereby maintaining patient confidentiality. This approach is especially suitable in dermatology, where data heterogeneity across hospitals can significantly affect the model's robustness.

Another central axis is ensuring that the model works robustly across dermatoscopes, smartphones, and portable imaging devices. DermaScanAI, with proper calibration and robustness analysis, can be incorporated into mobile diagnostic systems to potentially aid early diagnosis at the primary care level or in remote areas.

As a long-term outcome, DermaScanAI can mature into a comprehensive clinical decision support system by incorporating additional data modalities, including clinical images, histopathology images, sensitive patient data, and longitudinal patient information. The integration of visual and textual information would facilitate more comprehensive lesion evaluation, aid in risk stratification, and be beneficial to dermatologists in their diagnostic and management decisions. Therein, these future extensions set the stage for a scalable, clinically impactful dermatological support framework powered by AI potential via DSAI.

Discussion

Skin cancer is one of the most common types of cancer in the world today, making early detection and precise classification critical to improving survival rates and clinical outcomes. Although the automated analysis of dermatological images has advanced considerably, classical convolutional neural network (CNN) models, spatial inconsistencies, and class imbalances, which are characteristic of dermoscopic datasets, remain challenges. Most approaches lack attention mechanisms to highlight critical regions of lesions or cross-channel dependencies, which may result in suboptimal performance in skin cancer subtype classification, where many subtypes appear visually similar.

To address those issues, we propose a novel multi-scale attention-based hybrid DL framework that combines a Transformer block and Squeeze-and-Excitation (SE) blocks with a Convolutional Neural Network (CNN) backbone. This architecture presents several contributions: (i) a multi-scale CNN module that extracts a range of spatial features in a hierarchical way from dermoscopic images, (ii) an embedded SE block that recalibrates feature maps modelling inter-channel relationships, (iii) a Transformer module that extracts global contextual information, allowing the model to discriminate subtle patterns of lesions better. This combined design addresses fundamental flaws found in previous models that could not be spatially focused or learn global representations.

Quantitative analysis on the HAM10000 dataset demonstrates that the proposed model achieves significantly better performance than baseline CNN-only, Transformer-only, and other state-of-the-art hybrid models, with an AUC of 0.956, a macro F1-score of 91.60%, and an all-accuracy of 93.71%, respectively. For class-wise performance metrics (precision and recall), the model is robust to data imbalance. The model's robustness has significant implications for reducing false negatives and improving class separation, particularly between malignant classes such as melanoma (Fig. 8; Table 4). In addition, since Grad-CAM produces an overlay on the images, the interpretability findings, especially over clinically relevant lesion regions, further complement the model by introducing the necessary element of transparency and prospects for clinical adoption.

Combining global context and local attention, their work achieves higher accuracy than state-of-the-art methods. We propose a solution that scales well and is interpretable, which is consistent with the requirements of diagnostic practice. Ultimately, the research will lead to AI-enabled decision-support systems for dermatologists to assist them in making accurate, timely dermatologic diagnoses of skin cancer.

Clinical relevance, validation, and limitations

SkinNet-HDA, a proposed framework to support dermatological workflows with real-world-deployed applications, such as helping clinicians with early screening of skin lesions and risk stratification from dermoscopic images. In routine clinical practice, a dermatologist uses visual inspection, combined with patient metadata (age, sex, and anatomical site). To produce an intuitive and clinically-anchored decision support, SkinNet-HDA mimics this process by jointly optimally modelling image-driven lesion attributes and clinical metadata. Explainable Mechanisms can further increase trust, allowing clinicians to validate whether model attention aligns with clinically relevant lesion areas.

While these are strengths of the study, there are also significant limitations. In the first place, the primary assessment is performed on the HAM10000 dataset, which, although popular, suffers from dataset homogeneity regarding acquisition protocol and lesion appearance. This may restrict direct generalisation to images acquired with other dermatoscopes, other mobile devices, or in more heterogeneous clinical settings. Secondly, model performance is necessarily dependent on image quality, and some expected conditions, such as blur, occlusion, non-HMI lighting variation, and hair artefacts, can affect the certainty of predictions. Third, a domain shift is possible if the model is deployed to a different institution with different patient demographics, imaging standards, or annotation practices.

Future efforts will focus on multi-institutional validation, leveraging datasets obtained across institutions and care settings to mitigate these limitations and facilitate timely progression toward clinical translation. We will compare several prospective physician–AI studies by examining diagnostic concordance and the added value of AI-assisted decision-making. We will also investigate robustness under different image acquisition conditions and across device heterogeneity to ensure the reliable deployment of our results. Such steps are crucial to close the gap between experimental performance and safe confidence for clinical use. Section 6.2 provides a detailed discussion of the limitations encountered throughout the modelling and evaluation process, with recommendations for improving both development and deployment.

Limitations of the study

Despite a deep learning framework achieving promising results on the HAM10000 dataset for a set of skin cancer lesions, some limitations should be noted. First, the dataset suffers from an imbalance in pathological classifications, with fewer lesion types (e.g., vascular lesions and dermatofibroma). We can observe that performance has improved across all classes, even minority ones, but without evaluating on balanced or diverse datasets, we cannot say it generalises. Second, the training and evaluation images are collected from a dermoscope platform under stable conditions. Reducing the applicability of the framework in the clinical setting due to variations in image quality, lighting, and devices. Third, while the model's interpretability relies heavily on Grad-CAM visualisations, it is limited in some respects because the visual cues it generates may not be subtle to dermatologists. More robust embedding of explainable AI methods could further promote clinical trust and usability. This is similar to the exclusion of metadata (age, location, history) that can provide important context for the diagnosis, which is not included in the current study. Finally, the hybrid architecture (CNN + Transformer + SE blocks) increases the computational burden, making it difficult to deploy on resource-constrained devices. These limitations highlight possible avenues for improving the generalisation, interpretability, and deployment efficiency of decoding models in the years to come.

Conclusion and future work

In this paper, we propose a multi-scale attention-based deep learning framework that combines convolutional and Transformer modules and employs a squeeze-and-excitation (SE) step for accurate skin cancer lesion classification from extracted dermoscopic images. Our proposed architecture exploits CNNs' ability to extract local features and Transformer blocks' global contextual modelling, along with channel-wise attention from SE units, achieving competitive classification performance on the HAM10000 dataset across 7 classes. The model is evaluated across a plethora of tasks, from class-wise statistics to confusion matrices and ROC curves, demonstrating its robustness and discriminative ability. Grad-CAM visualisations further enhance interpretability and the model's potential for clinical adoption. It overcomes some of the limitations reflected in previous studies: mainly the limited research on multi-scale feature fusion, the underuse of attention mechanisms, and the lack of generalizability to their individual imbalanced lesion classes. However, a few challenges, such as real-world generalisation and class imbalance, and the lack of interpretability, remain and are discussed in Sect. 6.1. Future work will continue to improve external validity by using heterogeneous cohorts across clinical contexts. Increasing the specificity of a diagnosis can be achieved by adding more patient metadata (age, lesion history, and anatomical location). Finally, Grad-CAM, coupled with more advanced explainable AI methods, will increase clinical confidence and improve informed decision support. And finally, we will explore the techniques for model compression and the deployment of the framework on edge devices for real-time, resource-efficient clinical QSM applications. This is a substantial step toward AI-assisted dermatological diagnosis and supports further efforts to integrate AI into clinical practice reliably.

Data availability

The datasets used in this study are publicly available. Dermoscopic images were obtained from the HAM10000 dataset, which is accessible via the ISIC Archive ([https://www.isic-archive.com](https:/www.isic-archive.com)). All code related to data preprocessing, model training, evaluation, and explainability analysis for DermaScanAI is publicly available at [https://github.com/krishmu/DermaScanAI-SkinNetHDA](https:github.com/krishmu/DermaScanAI-SkinNetHDA). No proprietary or restricted data were used in this study.

Code availability

The code is available from the corresponding author and can be given on request.

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References

1. Tahir, M. et al. DSCC_Net: Multi-classification deep learning models for diagnosing of skin cancer using dermoscopic images. *Cancers* **15**(7), 2179 (2023).
2. Bechelli, S. & Delhommelle, J. Machine learning and deep learning algorithms for skin cancer classification from dermoscopic images. *Bioengineering* **9** (3), 97 (2022).
3. Grignaffini, F. et al. Machine learning approaches for skin cancer classification from dermoscopic images: A systematic review. *Algorithms* **15**(11), 438 (2022).
4. Nawaz, K. et al. Skin cancer detection using dermoscopic images with convolutional neural network. *Sci. Rep.* **15**(1), 7252 (2025).
5. Naeem, A., Anees, T., Fiza, M., Naqvi, R. A. & Lee, S. W. SCDNet: A deep learning-based framework for the multiclassification of skin cancer using dermoscopy images. *Sensors* **22**(15), 5652 (2022).
6. Shetty, B. et al. Skin lesion classification of dermoscopic images using machine learning and convolutional neural network. *Sci. Rep.* **12**(1), 18134 (2022).
7. Abunadi, I. & Senan, E. M. Deep learning and machine learning techniques of diagnosis dermoscopy images for early detection of skin diseases. *Electronics* **10** (24), 3158 (2021).
8. Ali, M. S., Miah, M. S., Haque, J., Rahman, M. M. & Islam, M. K. An enhanced technique of skin cancer classification using deep convolutional neural network with transfer learning models. *Mach. Learn. Appl.* **5**, 100036 (2021).

9. Wu, Y. et al. Skin cancer classification with deep learning: A systematic review. *Front. Oncol.* **12**, 893972 (2022).
10. Shah, S. A. H. et al. Explainable AI-based skin cancer detection using CNN, particle swarm optimization and machine learning. *J. Imaging.* **10** (12), 332 (2024).
11. Giavina-Bianchi, M. et al. Explainability agreement between dermatologists and five visual explanations techniques in deep neural networks for melanoma AI classification. *Front. Med.* **10**, 1241484 (2023).
12. de Souza, L. A. Jr. et al. Convolutional neural networks for the evaluation of cancer in Barrett's esophagus: Explainable AI to lighten up the black-box. *Comput. Biol. Med.* **135**, 104578 (2021).
13. Nigar, N., Umar, M., Shahzad, M. K., Islam, S. & Abalo, D. A deep learning approach based on explainable artificial intelligence for skin lesion classification. *IEEE Access* **10**, 113715–113725. https://doi.org/10.1109/ACCESS.2022.3217217 (2022).
14. Ahmad, N. et al. A novel framework of multiclass skin lesion recognition from dermoscopic images using deep learning and explainable AI. *Front. Oncol.* **13**, 1151257 (2023).
15. Nematzadeh, H., García-Nieto, J., Navas-Delgado, I. & Aldana-Montes, J. F. Ensemble-based genetic algorithm explainer with automized image segmentation: A case study on melanoma detection dataset. *Comput. Biol. Med.* **155**, 106613 (2023).
16. Yasir, S. M. & Kim, H. Attention-based deep feature aggregation network for skin lesion classification. *Electronics* **14**(12), 2364 (2025).
17. Shaik, A. et al. An attention based hybrid approach using CNN and BiLSTM for improved skin lesion classification. *Sci. Rep.* **15**(1), 15680 (2025).
18. Alahmadi, M. D. Multiscale attention U-net for skin lesion segmentation. *IEEE Access* **10**, 59145–59154. https://doi.org/10.1109/ACCESS.2022.3179390 (2022).
19. Liu, S., Zhuang, Z., Zheng, Y. & Kolmanič, S. A VAN-based multi-scale cross-attention mechanism for skin lesion segmentation network. *IEEE Access* **11**, 81953–81964. https://doi.org/10.1109/ACCESS.2023.3298826 (2023).
20. Wang, Y. et al. Adversarial multimodal fusion with attention mechanism for skin lesion classification using clinical and dermoscopic images. *Med. Image Anal.* **81**, 102535 (2022).
21. Gulzar, Y. & Khan, S. A. Skin lesion segmentation based on vision transformers and convolutional neural networks—a comparative study. *Appl. Sci.* **12**(12), 5990 (2022).
22. Abdelhalim, I. S. A., Mohamed, M. F. & Mahdy, Y. B. Data augmentation for skin lesion using self-attention based progressive generative adversarial network. *Expert Syst. Appl.* **165**, 113922 (2021).
23. Hossain, M. M., Hossain, M. M., Arefin, M. B., Akhtar, F. & Blake, J. Combining state-of-the-art pre-trained deep learning models: A noble approach for skin cancer detection using max voting ensemble. *Diagnostics* **14** (1), 89 (2023).
24. Jaiyeoba, O., Jaiyeoba, O., Ogbuju, E. & Oladipo, F. AI-based detection techniques for skin diseases: A review of recent methods, datasets, metrics, and challenges. *J. Future Artif. Intell. Technol.* **1**(3), 318–336 (2024).
25. Nancy, V. A. O., Prabhavathy, P., Arya, M. S. & Ahamed, B. S. Comparative study and analysis on skin cancer detection using machine learning and deep learning algorithms. *Multimedia Tools Appl.* **82** (29), 45913–45957 (2023).
26. Ghosh, H., Rahat, I. S., Mohanty, S. N., Ravindra, J. V. R. & Sobur, A. A study on the application of machine learning and deep learning techniques for skin cancer detection. *Int. J. Comput. Syst. Eng.* **18**(1), 51–59 (2024).
27. Gamil, S., Zeng, F., Alrifaey, M., Asim, M. & Ahmad, N. An efficient AdaBoost algorithm for enhancing skin cancer detection and classification. *Algorithms* **17** (8), 353 (2024).
28. Midasala, V. D. et al. MFEUsLNet: Skin cancer detection and classification using integrated AI with multilevel feature extraction-based unsupervised learning. *Eng. Sci. Technol. Int. J.* **51**, 101632 (2024).
29. Mahmoud, N. M. & Soliman, A. M. Early automated detection system for skin cancer diagnosis using artificial intelligent techniques. *Sci. Rep.* **14**(1), 9749 (2024).
30. Natha, P. et al. Boosting skin cancer diagnosis accuracy with ensemble approach. *Sci. Rep.* **15**(1), 1290 (2025).
31. Saeed, M., Naseer, A., Masood, H., Rehman, S. U. & Gruhn, V. The power of generative AI to augment for enhanced skin cancer classification: A deep learning approach. *IEEE Access* **11**, 130330–130344. https://doi.org/10.1109/ACCESS.2023.3332628 (2023).
32. Musthafa, M. M. et al. Enhanced skin cancer diagnosis using optimized CNN architecture and checkpoints for automated dermatological lesion classification. *BMC Med. Imaging* **24**, 201. https://doi.org/10.1186/s12880-024-01356-8 (2024).
33. Gururaj, H. L., Manju, N., Nagarjun, A., Aradhya, V. N. M. & Flammini, F. Deepskin: A deep learning approach for skin cancer classification. *IEEE Access* **11**, 50205–50214. https://doi.org/10.1109/ACCESS.2023.3274848 (2023).
34. Nawaz, M. et al. Skin cancer detection from dermoscopic images using deep learning and fuzzy k-means clustering. *Microsc. Res. Tech.* **85**(1), 339–351 (2022).
35. Kaur, R., GholamHosseini, H., Sinha, R. & Lindén, M. Melanoma classification using a novel deep convolutional neural network with dermoscopic images. *Sensors* **22** (3), 1134 (2022).
36. Venugopal, V., Raj, N. I., Nath, M. K. & Stephen, N. A deep neural network using modified EfficientNet for skin cancer detection in dermoscopic images. *Decis. Anal. J.* **8**, 100278 (2023).
37. Mehr, R. A. & Ameri, A. Skin cancer detection based on deep learning. *J. Biomed. Phys. Eng.* **12**(6), 559 (2022).
38. Naeem, A. et al. SNC_Net: Skin cancer detection by integrating handcrafted and deep learning-based features using dermoscopy images. *Mathematics* **12**(7), 1030 (2024).
39. Grignaffini, F., De Santis, E., Frezza, F. & Rizzi, A. An XAI approach to melanoma diagnosis: Explaining the output of convolutional neural networks with feature injection. *Information* **15**(12), 783 (2024).
40. Mridha, K., Uddin, M. M., Shin, J., Khadka, S. & Mridha, M. F. An interpretable skin cancer classification using optimized convolutional neural network for a smart healthcare system. *IEEE Access* **11**, 41003–41018. https://doi.org/10.1109/ACCESS.2023.3269694 (2023).
41. Gamage, L., Isuranga, U., Meedeniya, D., De Silva, S. & Yogarajah, P. Melanoma skin cancer identification with explainability utilizing mask guided technique. *Electronics* **13** (4), 680 (2024).
42. Ali, A., Shahbaz, H. & Damaševičius, R. XCViT: Improved vision transformer network with fusion of CNN and Xception for skin disease recognition with explainable AI. *Comput. Mater. Contin.* https://doi.org/10.32604/cmc.2025.059301 (2025).
43. Dagnaw, G. H., El Mouhtadi, M. & Mustapha, M. Skin cancer classification using vision transformers and explainable artificial intelligence. *J. Med. Artif. Intell.* https://doi.org/10.21037/jmai-24-6 (2024).
44. Ullah, M. A. & Zia, T. Hybrid interpretable deep learning framework for skin cancer diagnosis: integrating radial basis function networks with explainable AI. *arXiv preprint arXiv:2501.14885*. (2025).
45. Liu, X. et al. Predicting skin cancer risk from facial images with an explainable artificial intelligence (XAI) based approach: A proof-of-concept study. *EClinicalMedicine* https://doi.org/10.1016/j.eclinm.2024.102550 (2024).
46. Behara, K., Bhero, E. & Agee, J. T. Grid-based structural and dimensional skin cancer classification with self-featured optimized explainable deep convolutional neural networks. *Int. J. Mol. Sci.* **25**(3), 1546 (2024).
47. Chanda, T. et al. Dermatologist-like explainable AI enhances trust and confidence in diagnosing melanoma. *Nat. Commun.* **15**(1), 524 (2024).
48. Nie, T., Zhao, Y. & Yao, S. ELA-Net: An efficient lightweight attention network for skin lesion segmentation. *Sensors* **24**(13), 4302 (2024).
49. Zhuang, J. X., Cai, J., Zhang, J., Zheng, W. S. & Wang, R. Class attention to regions of lesion for imbalanced medical image recognition. *Neurocomputing* **555**, 126577 (2023).
50. Manzari, O. N., Kaleybar, J. M., Saadat, H. & Maleki, S. BEFUnet: A hybrid CNN-transformer architecture for precise medical image segmentation. *arXiv preprint arXiv:2402.08793*. (2024).

51. Takiddin, A., Schneider, J., Yang, Y., Abd-Alrazaq, A. & Househ, M. Artificial intelligence for skin cancer detection: Scoping review. *J. Med. Internet Res.* **23**(11), e22934 (2021).
52. Bhatt, H., Shah, V., Shah, K., Shah, R. & Shah, M. State-of-the-art machine learning techniques for melanoma skin cancer detection and classification: A comprehensive review. *Intell. Med.* **3**(03), 180–190 (2023).
53. Alzamili, A. H. & Ruhaiyem, N. I. R. A comprehensive review of deep learning and machine learning techniques for early-stage skin cancer detection: Challenges and research gaps. *J. Intell. Syst.* **34**(1), 20240381 (2025).
54. Behara, K., Bhero, E. & Agee, J. T. AI in dermatology: A comprehensive review into skin cancer detection. *PeerJ Comput. Sci.* **10**, e2530 (2024).
55. Rezk, E., Haggag, M., Eltorki, M. & El-Dakhakhni, W. A comprehensive review of artificial intelligence methods and applications in skin cancer diagnosis and treatment: Emerging trends and challenges. *Healthc. Anal.* **4**, 100259 (2023).
56. Melarkode, N., Srinivasan, K., Qaisar, S. M. & Plawiak, P. AI-powered diagnosis of skin cancer: A contemporary review, open challenges and future research directions. *Cancers* **15**(4), 1183 (2023).
57. Abou Ali, M., Dornaika, F., Arganda-Carreras, I., Ali, H. & Karaouni, M. Naturalize revolution: Unprecedented AI-driven precision in skin cancer classification using deep learning. *BioMedInformatics* **4**(1), 638–660 (2024).
58. Khater, T., Ansari, S., Mahmoud, S., Hussain, A. & Tawfik, H. Skin cancer classification using explainable artificial intelligence on pre-extracted image features. *Intell. Syst. Appl.* **20**, 200275 (2023).
59. Ravi, V. Attention cost-sensitive deep learning-based approach for skin cancer detection and classification. *Cancers* **14** (23), 5872 (2022).
60. Naveed, A. et al. AD-Net: Attention-based dilated convolutional residual network with guided decoder for robust skin lesion segmentation. *Neural Comput. Appl.* **36**(35), 22277–22299 (2024).
61. Pedro, R. & Oliveira, A. L. Assessing the impact of attention and self-attention mechanisms on the classification of skin lesions. In *2022 International Joint Conference on Neural Networks (IJCNN)* (pp. 1–8). IEEE. (2022), July.
62. Younas, S., Sargano, A. B., You, L. & Habib, Z. Attention-based inception-residual CNN: Skin cancer diagnosis with attention-based inception-residual CNN model. *Information* **16**(2), 120 (2025).
63. Noor, M. N., Haneef, F., Ashraf, I. & Masud, M. Enhanced skin disease classification via dataset refinement and attention-based vision approach. *Bioengineering* **12**(3), 275 (2025).
64. Sharma, R. R. et al. Comparative analysis of vision transformer and cnn architectures in medical image classification. In *International Conference on Sustainability Innovation in Computing and Engineering (ICSICE 2024)* (pp. 1343–1355). Atlantis Press. (2025), May.
65. Bougourzi, F., Dornaika, F., Distante, C. & Taleb-Ahmed, A. D-TrAttUnet: Toward hybrid CNN-transformer architecture for generic and subtle segmentation in medical images. *Comput. Biol. Med.* **176**, 108590 (2024).
66. Kim, J. W., Khan, A. U. & Banerjee, I. Systematic review of hybrid vision transformer architectures for radiological image analysis. *J. Imaging Inform. Med.* https://doi.org/10.1007/s10278-024-01322-4 (2025).
67. Wen, D., Soltan, A., Trucco, E. & Matin, R. N. From data to diagnosis: Skin cancer image datasets for artificial intelligence. *Clin. Exp. Dermatol.* **49**(7), 675–685 (2024).
68. Nguyen, A. T. P., Shak, M. S. & Al-Imran, M. Advancing early skin cancer detection: A comparative analysis of machine learning algorithms for melanoma diagnosis using dermoscopic images. *International Journal of Medical Science and Public Health Research* **5**(12), 119–133 (2024).
69. Djoumessi, K., Mensah, S. O. & Berens, P. A Hybrid Fully Convolutional CNN-Transformer Model for Inherently Interpretable Medical Image Classification. *arXiv preprint arXiv:2504.08481*. (2025).
70. Hu, C., Cao, N., Zhou, H. & Guo, B. Medical image classification with a hybrid SSM model based on CNN and transformer. *Electronics* **13**(15), 3094 (2024).
71. Zeynali, A., Tinati, M. A. & Tazehkand, B. M. Hybrid CNN-Transformer architecture with Xception-based feature enhancement for accurate breast cancer classification. *IEEE Access* **12**, 189477–189493. https://doi.org/10.1109/ACCESS.2024.3516535 (2024).
72. Li, Z. et al. Tfcns: A cnn-transformer hybrid network for medical image segmentation. In *International conference on artificial neural networks* (pp. 781–792). Cham: Springer Nature Switzerland. (2022), September.
73. Lan, L. et al. Brau-net++: U-shaped hybrid cnn-transformer network for medical image segmentation. *arXiv preprint arXiv:2401.00722*. (2024).
74. Vindas, Y., Guépié, B. K., Almar, M., Roux, E. & Delachartre, P. An hybrid cnn-transformer model based on multi-feature extraction and attention fusion mechanism for cerebral emboli classification. In *Machine Learning for Healthcare Conference* (pp. 270–296). PMLR. (2022), December.
75. Singh, S. Computer-aided diagnosis of thoracic diseases in chest X-rays using hybrid cnn-transformer architecture. *arXiv preprint arXiv:2404.11843*. (2024).
76. Kuang, H. et al. Hybrid cnn-transformer network with circular feature interaction for acute ischemic stroke lesion segmentation on non-contrast ct scans. *IEEE Trans. Med. Imaging*. **43** (6), 2303–2316 (2024).
77. Gu, P., Zhang, Y., Wang, C. & Chen, D. Z. Convformer: Combining cnn and transformer for medical image segmentation. In *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)* (pp. 1–5). IEEE. (2023), April.
78. Bi, L. et al. Hybrid CNN-transformer network for interactive learning of challenging musculoskeletal images. *Comput. Methods Programs Biomed.* **243**, 107875 (2024).
79. Iqbal, S., Khan, T. M., Naqvi, S. S., Naveed, A. & Meijering, E. TBConvL-Net: A hybrid deep learning architecture for robust medical image segmentation. *Pattern Recognit.* **158**, 111028 (2025).
80. Banerjee, T. et al. Pyramidal attention-based T network for brain tumor classification: a comprehensive analysis of transfer learning approaches for clinically reliable and reliable AI hybrid approaches. *Sci. Rep.* **15**, 28669. https://doi.org/10.1038/s41598-025-11574-x (2025).
81. Banerjee, T., Singh, D. P., Kour, P., Swain, D. & Narayan, Y. Comparing bipartite convoluted and attention-driven methods for skin cancer detection: a review of explainable AI and transfer learning strategies. *Arch. Comput. Methods Eng.* https://doi.org/10.1007/s11831-025-10379-5 (2025).
82. Banerjee, T., Singh, D. P. & Kour, P. Advances in deep neural, transformer learning, and kernel-based methods for diabetic retinopathy detection: a comprehensive review. *Arch. Comput. Methods Eng.* https://doi.org/10.1007/s11831-025-10376-8 (2025).
83. Singh, D. P., Banerjee, T. & Mahajan, S. A comprehensive study on deep learning models for the detection of diabetic retinopathy using pathological images. *Arch. Comput. Methods Eng.* https://doi.org/10.1007/s11831-025-10315-7 (2025).
84. Sujini, G. N. Automated thyroid nodule classification in ultrasound imaging using Vision Transformer and WGAN-GP hybrid model. *Sci. Rep.* https://doi.org/10.1038/s41598-025-24651-y (2025).
85. Tschandl, P., Rosendahl, C. & Kittler, H. The HAM10000 dataset: A large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Sci. Data* **5**, 180161. https://doi.org/10.1038/sdata.2018.161 (2018).
86. He, X. et al. Deep metric attention learning for skin lesion classification in dermoscopy images. *Complex Intell. Syst.* **8**, 1487–1504. https://doi.org/10.1007/s40747-021-00587-4 (2022).

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Consent for publication
The authors give consent for their publication.

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